Oncology Basics, Genitalia, DSD, Gender & Ageing Rapid Revision

Campbell-Walsh-Wein · Chapters 24, 25, 55, 58–60, 65, 120, 126–128

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 5h 19min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 60* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
24	Basic Principles of Immunology and Immunotherapy in Urologic Oncology	18 min
25	Molecular Genetics and Cancer Biology	30 min
55	Management of Abnormalities of the External Genitalia in Males	66 min
58	Management of Abnormalities of the Genitalia in Girls	22 min
59	Differences of Sexual Development	36 min
60	Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations	44 min
65	Adolescent and Transitional Urology	14 min
120	Aging and Geriatric Urology	21 min
126	Management and Considerations in Transgender and Nonbinary Individuals	22 min
127	Male-to-Female Gender-Affirming Surgery: Vaginoplasty	22 min
128	Special Urologic Considerations in Transmasculine Individuals	16 min
★	Putting it together: genitalia, DSD, gender and special populations	8 min
	Total (22408 words, at 70 words/min)	**5h 19min**

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

Reading time ≈ 18 min (slow pace)

1 · INNATE AND ADAPTIVE IMMUNITY

- **Innate**: evolutionarily older, rapid, pattern-based. **Adaptive**: specific, with memory (stronger response on re-encounter).
- Danger sensing: pathogen-associated molecular patterns (PAMPs) bind **Toll-like receptors (TLRs)**. Bacille Calmette-Guérin (BCG) cell-wall peptidoglycans are canonical PAMPs → **TLR2** on bladder-wall innate cells.

Cell	Arm	Role
Urothelial cell	Sentinel	Type I interferons, chemokines on danger signals
Macrophage, neutrophil	Innate	Engulf pathogens; amplify response
Dendritic cell (DC)	Bridge	Carries antigen to lymph node, presents to T cells
CD4 T cell	Adaptive	"Helper" for CD8 killing or B-cell antibody
CD8 T cell	Adaptive	"Killer" — serial, specific, apoptosis-inducing lysis
Regulatory T cell (Treg)	Adaptive	**CD4 subset, marker FoxP3**; suppresses responses — detrimental in cancer

Cytokines and T-helper polarisation

- Type I interferons (IFN-α/β) upregulate class I major histocompatibility complex (MHC) and antigen processing. Chemokine CXCL8 (interleukin-8, IL-8) from urothelium = powerful neutrophil attractant.
- **IL-12 → T-helper 1 (Th1)**; **IL-4 → Th2**. Th1 outputs IL-2, tumour necrosis factor-α (TNF-α), IFN-γ → cellular, cytotoxic, antitumour → **successful BCG**. Th2 outputs IL-4, IL-5, IL-10, IL-13 → humoral, chronic inflammation → **BCG failure**.
- Raised serum IL-2 after BCG predicts a favourable outcome.

Dendritic cells and T cells

ALGORITHM
Danger signal (TNF-α, TLR ligand) → DC stops antigen uptake → upregulates MHC and costimulatory B7-1/B7-2 → migrates via afferent lymphatics (CCL21 sensed by CCR7) → activates CD4, CD8, B cells in node

- Activation needs **signal 1** (T-cell receptor [TCR] + MHC–peptide) and **signal 2** (CD28 + CD80/CD86 costimulation).

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; ↑ by inflammation, IFN	Antigen-presenting cells (APCs)
Peptide	~9 amino acids	13–25 amino acids

2 · IMMUNE EDITING AND ESCAPE

- Three Es: **Elimination** → **Equilibrium** (may last lifetime) → **Escape** (clinical tumour). Escape: ↓ tumour antigen, ↓ MHC, ↑ Tregs.
- **Adaptive immune resistance**: CD8 cells recognise mutant antigen → secrete IFN-γ → IFN-γ induces programmed death ligand-1 (PD-L1) on tumour/myeloid cells → binds programmed death-1 (PD-1) on the same CD8 cells → effector function curtailed.

> **KEY POINT** PD-L1 is largely induced by the immune attack itself — a marker of an ongoing but suppressed response, not a fixed tumour trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating costimulation
Cytotoxic T-lymphocyte antigen-4 (CTLA-4)	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
Lymphocyte activation gene-3 (LAG-3)	MHC class II	Inhibitory

3 · INFLAMMATION AND THE GENITOURINARY (GU) TUMOURS

	Bladder	Kidney	Prostate
Inflammation → cancer	Strong: *Schistosoma haematobium* (squamous), chronic urinary tract infection (UTI), catheters, cyclophosphamide	Weak (smoking, obesity)	Probable: peripheral-zone inflammation; proliferative inflammatory atrophy (PIA) beside high-grade prostatic intraepithelial neoplasia
CD8 infiltration	Better outcome (muscle-invasive)	**Worse outcome — renal paradox**	CD8 cells may be suppressive
Dominant restraint	PD-L1, ↑ with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

- Chronic Th2/Treg/M2-macrophage inflammation promotes cancer; acute Th1/CD8 response eliminates it.

4 · BCG IN BLADDER CANCER

ALGORITHM

Instil 1–5 × 10^8 bacilli → fraction adheres via **fibronectin attachment protein** → TLR2/TLR4 on urothelium → IL-8 + TNF-α → neutrophil wave (hours) → DC activation → CD4 Th1 response in nodes (weeks) → memory

- Memory implied by ~**50%** of non-muscle-invasive bladder cancer (NMIBC) in durable remission.

Nadofaragene firadenovec (BCG-unresponsive NMIBC)

- IFN-α2b protein is not durable (short dwell) → non-replicating adenovirus carrying the IFN-α2b gene + **Syn3** (overcomes the antiadherence barrier).
- 3-monthly up to 12 months. Carcinoma in situ (CIS) complete response (CR) 53.4%. Food and Drug Administration (FDA) 2022 — **first gene therapy** for BCG-unresponsive NMIBC with CIS.

5 · CANCER VACCINES

- Components: **adjuvant** (activates DCs; granulocyte-macrophage colony-stimulating factor [GM-CSF] is the recurring one) + tumour-overexpressed **antigen**.
- **Sipuleucel-T**: autologous peripheral blood mononuclear cells incubated with PA2024 (prostatic acid phosphatase [PAP]–GM-CSF fusion), intravenous every 2 weeks × 3; asymptomatic/minimally symptomatic metastatic castration-resistant prostate cancer (mCRPC). Overall survival (OS) **+4.1 months** but **no effect on time to progression**; first FDA-approved solid-tumour vaccine.

- All other GU phase III vaccines failed: IMA901 and AGS-003 (renal cell carcinoma [RCC]), PROSTVAC-VF (prostate).

6 · CHECKPOINT BLOCKADE

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder (urothelial carcinoma)

- Single-agent first-line pembrolizumab (KEYNOTE-052) is restricted to cisplatin-ineligible with combined positive score (CPS) **≥10**, or any-platinum-ineligible — CPS <10 did worse than carboplatin chemotherapy.
- Atezolizumab and durvalumab withdrawn (failed confirmatory trials); adjuvant atezolizumab (IMvigor010) negative — why nivolumab succeeded is unclear (design vs PD-1 vs PD-L1 target).
- **Enfortumab vedotin (EV)**: antibody–drug conjugate (ADC), anti-nectin-4 + monomethyl auristatin E. EV + pembrolizumab: objective response rate (ORR) 68% in cisplatin-ineligible (EV-103). †EV-302 (phase III): EV + pembrolizumab roughly doubled progression-free survival (PFS) and OS vs platinum chemotherapy → new first-line standard regardless of cisplatin eligibility.

Kidney

- CTLA-4 is overexpressed on Tregs; PD-1 on CD8 TILs → combined blockade is synergistic.
- Vascular endothelial growth factor (VEGF) is immunosuppressive (impairs DC maturation, T-cell entry); tyrosine kinase inhibitor (TKI) VEGF blockade normalises vessels → more CD8 cells → rationale for TKI + checkpoint inhibitor.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- Pembrolizumab in mCRPC: prostate-specific antigen (PSA) response <10% — tumour cells rarely express PD-L1. Ipilimumab: two phase III trials, no OS benefit.
- Exception: **microsatellite instability-high (MSI-H)/mismatch repair-deficient (dMMR)** (1.2–12%) → many neoantigens → pembrolizumab second line or beyond. 2017 tumour-agnostic approval — **first based on a biomarker, not site**.
- Enzalutamide raises PD-L1, but atezolizumab + enzalutamide (IMbassador250) gave no OS benefit.
- †Immune-related adverse events (colitis, hepatitis, pneumonitis, endocrinopathies): hold the drug and give corticosteroids for grade ≥2; endocrinopathies need lifelong hormone replacement.

> **KEY POINT** Every successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, immunogenic cell death, VEGF inhibition) or removes a brake (PD-1, CTLA-4). Failures cluster where the brake is not limiting — PD-L1-negative, Treg-dominated prostate cancer.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay

Trial	Design	Result	Take-home
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only anti-PD-1/PD-L1 approved on phase III second-line data
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk muscle-invasive	Disease-free survival 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, metastatic RCC after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab 1 year vs placebo, high-risk clear-cell RCC	Recurrence/death ↓32%	Adjuvant standard

8 · MCQ ONE-LINERS

1. BCG PAMPs signal through TLR2 (and TLR4).
2. BCG adheres via fibronectin attachment protein.
3. Th1 (IL-2, IFN-γ, TNF-α) = BCG success; Th2 = failure.
4. IL-12 drives Th1; IL-4 drives Th2.
5. Treg: CD4 cell marked by FoxP3.
6. CD8 recognises class I MHC, ~9-amino-acid peptides.
7. Immune editing: elimination, equilibrium, escape.
8. PD-L1 is induced by IFN-γ from infiltrating CD8 cells.
9. Renal paradox: high CD8 density predicts worse RCC outcome.
0. Sipuleucel-T antigen: prostatic acid phosphatase.
1. Sipuleucel-T prolongs OS without delaying progression.
12. Nadofaragene: adenoviral IFN-α2b with Syn3; first gene therapy for NMIBC.
13. CTLA-4 is mainly on Tregs; PD-1 on CD8 TILs.
14. Avelumab: PD-L1 inhibitor, maintenance after platinum.
15. First-line pembrolizumab monotherapy needs CPS ≥10 if cisplatin-ineligible.
16. Enfortumab vedotin targets nectin-4.
17. First tumour-agnostic biomarker approval: pembrolizumab for MSI-H/dMMR.
18. Anti-PD-1 fails in most prostate cancer: tumour PD-L1-negative.
19. Nivolumab + ipilimumab: first line intermediate/poor-risk RCC.
20. Highest ORR among TKI combinations: pembrolizumab + lenvatinib.

25 Molecular Genetics and Cancer Biology

Reading time ≈ 30 min (slow pace)

1 · HALLMARKS AND BASIC MOLECULAR GENETICS

- Carcinogenesis is multistep over years. Attributes of the malignant cell: genetic instability, autonomous growth, insensitivity to antiproliferative signals, resistance to apoptosis, unlimited division, angiogenesis, **local invasion (the one feature that uniquely separates malignant from benign)**, immune evasion; many acquire metastasis.
- Deoxyribonucleic acid (DNA) → messenger ribonucleic acid (mRNA) → protein. Pre-mRNA undergoes 5′ capping, 3′ polyadenylation and splicing (introns out, exons joined).
- **Alternative splicing** lets one gene encode several proteins: androgen receptor (AR) splice variants lacking the ligand-binding domain (e.g. AR-V7) are active without ligand → castration resistance.

	Somatic testing	**Germline testing**
Detects	Acquired, tumour-confined mutations	Inherited variants in every cell
Sample	Tumour tissue	Blood or saliva
Use	Targetable mutations, resistance, heterogeneity	Hereditary syndromes, surveillance, preventive surgery, family cascade

2 · GENE ALTERATIONS: TUMOUR SUPPRESSORS AND ONCOGENES

Category	**Consequence**
Point mutation	Silent, missense (amino-acid substitution), nonsense (premature stop)
Insertion/deletion (indel)	Frameshift → usually non-functional protein
Chromosomal rearrangement	Inversions, translocations → deletions, amplification, gene fusions; chromothripsis

	Tumour suppressor	**Oncogene**
Normal role	Brake on growth; repair	Accelerator (proto-oncogene)
Cancer change	Loss of function — usually **both alleles**	Gain of function — **one allele**
Mechanism	Homozygous deletion; loss + mutation; two mutations; loss + promoter methylation	Activating mutation, amplification, rearrangement/fusion
Genitourinary (GU) examples	RB1, TP53, CDKN1B, VHL, PTEN	MYC, MET, ERBB/epidermal growth factor receptor (EGFR), TMPRSS2–ERG

- **Knudson two-hit hypothesis** (retinoblastoma): familial cases — often bilateral — inherit one bad allele, so one somatic hit suffices.
- Exceptions: **dominant negative** (mutant protein disables wild-type partner in multimers, e.g. TP53); **haploinsufficiency** (one allele loss lowers protein enough, e.g. p27/CDKN1B).
- MYC amplified in prostate and bladder cancer; MET germline activation causes hereditary papillary renal cell carcinoma (RCC).

3 · CELL CYCLE

- G0 (reversible quiescence) → G1 → S (DNA synthesis) → G2 → M (mitosis); ~24 h, unidirectional.
- Cyclin-dependent kinases (CDKs) are active only with their phase-specific **cyclin**; cyclins are degraded by the proteasome after each phase. Also regulated by CDK phosphorylation and CDK inhibitors (CDKIs).

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins → nuclear envelope breakdown

Control point	Trigger	Mechanism
G1/S	Mitogens, nutrients, oxygen, DNA damage	p21 (CDKN1A), 14-3-3σ inactivate G1 CDKs; TP53 → apoptosis if unrepaired
G2/M	DNA damage (e.g. radiation)	ATM → CHEK1/CHEK2 → Cdc25 sequestered; TP53 induces p21, GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex

- Spindle-checkpoint failure → non-disjunction → **aneuploidy**, near-universal in solid tumours.

	Cip/Kip	INK4
Members	p21, p27, p57	p15, p16, p18, p19
Range	Broad — many cyclin–CDK complexes	CDK4/6 only → block RB1 phosphorylation

RB1 — guardian of G1/S

ALGORITHM
Hypophosphorylated RB1 binds E2F + histone deacetylases (HDACs) → S-phase genes silent → cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1 → E2F released → S-phase genes transcribed → restriction point passed (irreversible commitment)

- Disabled by RB1 loss, cyclin D amplification or CDKI loss — altered in almost all cancers.

TP53 — guardian of the genome

- DNA damage (ATM/ATR → CHK1/CHK2) or oncogene stress (ARF blocks MDM2) → active TP53 → arrest, repair, senescence or apoptosis. **MDM2** is its negative regulator.
- Most commonly mutated gene in cancer (>half of malignancies); mediates both G1/S and G2/M arrest.

> **KEY POINT** Cancer must break both RB1 (entry) and TP53 (integrity) — which is why these two pathways are lost in almost every tumour.

4 · DNA DAMAGE AND REPAIR

- DNA damage response (DDR): arrest first (ATM, ATR activate TP53, CHK1), then repair.
- **Loeb mutator hypothesis**: early loss of replication fidelity → hypermutable tumour that rapidly selects resistant clones.

Pathway	Lesion	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky adducts, cross-links (ultraviolet, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
Base excision repair (BER)	Single-base damage, single-strand breaks (oxidation, alkylation, deamination)	MUTYH, OGG1, PARP1/2	OGG1 variants → RCC
Mismatch repair (MMR)	Mispairs, insertion/deletion loops at repeats	MSH2/MSH6, MLH1/PMS2	~100-fold ↑ mutation rate; Lynch
Homologous recombination (HR)	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate

Pathway	Lesion	Key proteins	Cancer link
Non-homologous end joining (NHEJ)	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- DSB: ATM activates → phosphorylates TP53, CHEK1/2 and H2AX → **γH2AX** foci recruit repair factors. Unrepaired DSBs → apoptosis — the basis of radiotherapy and many cytotoxics.
- **HR** uses the sister chromatid (only after S, in G2) → precise. **NHEJ** ligates ends without template in any phase → error-prone, translocations.

> **KEY POINT** Repair defects are therapeutic targets: HR loss → poly(ADP-ribose) polymerase (PARP) inhibitor (synthetic lethality); MMR loss → immunotherapy; ERCC2 loss → cisplatin sensitivity.

5 · APOPTOSIS

	Apoptosis	Necrosis
Nature	Programmed, energy-requiring, irreversible	Acute injury
Inflammation	Little	Potent

- Caspases: initiators 8, 9, 10 → executioners 3, 6, 7 (activated by cleavage).

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, loss of survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak → cytochrome c release → Apaf-1 apoptosome → caspase-9	FADD → procaspase-8 → caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; inhibitors of apoptosis proteins (IAPs)	FLIP
TP53	Dependent (transactivates Bax)	**Independent** → target when TP53 mutated

- Bid (cleaved by caspase-8) links extrinsic to intrinsic. Chemotherapy and radiotherapy largely need TP53 to kill, so TP53 loss confers resistance.

6 · SENESCENCE, TELOMERES, AUTOPHAGY

- **Senescence**: viable but permanently arrested (p16, p21, p27); marker senescence-associated β-galactosidase; secretes a senescence-associated secretory phenotype (SASP) — immunostimulatory but potentially pro-tumour. Androgen-deprivation therapy (ADT) induces therapy-induced senescence in some prostate cancers; SASP may explain ADT-associated inflammation.
- **Telomeres** lose ~100 bp per division = "mitotic clock". Shortening → DDR → senescence (barrier 1) → if checkpoints lost, crisis with chromosomal instability (barrier 2) → **telomerase** activation → immortalisation.
- Telomerase (reverse transcriptase) is high in **85–90%** of tumours; normally only in testis, endometrium, gut, activated lymphocytes. Alternative lengthening of telomeres (ALT) is telomerase-independent (ATRX/DAXX loss).
- Short telomeres in premalignant bladder/prostate lesions (early, causal); in prostate cancer they predict recurrence.
- **Autophagy** (lysosomal self-digestion; driven via ULK1 → Beclin-1): suppresses early tumours (BECN1 loss in prostate cancer) but sustains established tumours under hypoxia and therapy. Chloroquine inhibits it; no clinical urological role yet.

7 · EPIGENETICS

- **Reversible** expression change without sequence change → biomarker and drug target.
- **DNA methylation**: 5-methylcytosine at CpG islands (upstream of ~40% of promoters). DNMT1 maintains; DNMT3A/3B methylate de novo. 5-methylcytosine deaminates to thymine → C→T

transitions.

	Promoter hypermethylation	Global hypomethylation
Effect	Silences suppressors (acts as a "second hit")	Genomic instability; oncogene/imprinted gene re-expression
GU examples	**GSTP1 in >90% of prostate cancer**; RASSF1A (up to 97% of bladder); p16	LINE-1 in bladder; IGF2 loss of imprinting in prostate

- **Histone marks**: activating H3K4me3 and acetylation; repressive H3K9me3, **H3K27me3 (laid down by EZH2)**. EZH2 ↑ in metastatic castration-resistant prostate cancer and predicts prostate-specific antigen (PSA) recurrence.
- Approved DNMT and HDAC inhibitors work in haematological cancers; limited in solid tumours.

Non-coding RNA (ncRNA)	Feature	GU example
Long ncRNA	>200 nucleotides	HOTAIR → castration resistance; MALAT1
MicroRNA (miRNA)	18–25 nucleotides; silences **many** mRNAs	**miR-371a-3p** in testis cancer
Small interfering RNA	Silences **one** mRNA	—

8 · HYPOXIA, VHL AND ANGIOGENESIS

ALGORITHM
Normoxia: hypoxia-inducible factor-α (HIF-α) prolyl-hydroxylated → bound by von Hippel-Lindau protein (VHL, E3 ubiquitin ligase) → proteasomal degradation. Hypoxia or VHL loss: HIF-α accumulates → dimerises with HIF-1β → vascular endothelial growth factor (VEGF), angiopoietin and glycolytic genes

- **Warburg effect**: aerobic glycolysis, best seen in clear cell RCC with constitutive HIF-1α/HIF-2α.
- Hypoxia also drives invasion and immunosuppression (myeloid suppressor cells, programmed death ligand-1 [PD-L1], blocked dendritic-cell maturation).

KEY POINT VHL loss = permanent pseudohypoxia; angiogenesis, Warburg metabolism, invasion and immunosuppression all follow — why clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Cancer stem cell model**: a small quiescent, resilient population regenerates the tumour (response then recurrence); **stochastic model**: any cell may survive and regrow.
- **Lineage plasticity**: under intense driver targeting, cells trans-differentiate to a lineage with new drivers (e.g. neuroendocrine prostate cancer).

9 · NEXT-GENERATION SEQUENCING (NGS)

Technique	Measures
Whole-exome sequencing (WES)	Coding mutations, indels, copy number — mutational landscape
RNA sequencing	Expression, splicing, fusion transcripts
Bisulfite sequencing	Methylation (unmethylated C → uracil)
Single-cell RNA sequencing	Per-cell heterogeneity

10 · GU TUMOUR GENOMICS

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half); androgen-driven promoter drives ERG
SPOP	Commonest mutated suppressor in primary tumours
TP53	Commonest alteration overall (more in advanced disease)
RB1, PTEN, TP53	Loss of ≥2 = aggressive variant prostate cancer, overlaps neuroendocrine
BRCA1/2	Germline + somatic hit → **PARP inhibitor** (approved)
ATM	Little PARP benefit — distinct from BRCA
CDK12	Tandem duplications; poor response to hormones, chemotherapy, PARP
MMR deficiency	Microsatellite instability → immunotherapy
AR amplification	Late, under AR-targeted pressure → ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** + VHL, PBRM1, BAP1, SETD2	VEGF tyrosine kinase inhibitors (TKIs); belzutifan (HIF-2α inhibitor)
Papillary (15–20%)	MET (7q) activation; fumarate hydratase (FH) loss in a subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mammalian target of rapamycin (mTOR) pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1+ → immunotherapy-responsive

- Clear cell: VHL mutation 60–70% + methylation 5–10%; **PBRM1** (~40%) better outcome; **BAP1** and **SETD2** (~10% each) worse; PBRM1 and BAP1 mutually exclusive.

Bladder (urothelial carcinoma, UC)

- High mutation load (APOBEC signature); TP53 most frequent; chromatin modifiers (KDM6A, ARID1A), FGFR3, PIK3CA, RB1.
- Consensus six subtypes: **basal/squamous largest (35%)**, luminal papillary (24%), neuroendocrine-like smallest (3%).
- **FGFR3** (15–20% advanced; up to 37% upper tract): luminal papillary, T-cell-poor, poor immunotherapy response → **erdafitinib**; test all advanced UC for FGFR.
- HER2 (ERBB2) 9–12% of muscle-invasive. DDR alterations (ERCC2) → cisplatin sensitivity, bladder-sparing trials.

Testis

- Low mutation rate; **12p gain** (†classically isochromosome 12p); KIT/KRAS mutations (seminoma); TP53 mutations rare. miR-371a-3p: >90% sensitivity and specificity.

Hereditary syndromes

Syndrome	Gene	GU tumour
Von Hippel-Lindau (autosomal dominant [AD])	VHL	Clear cell RCC (>70% lifetime); belzutifan
Hereditary papillary RCC (AD)	MET (activating)	Papillary type 1
Hereditary leiomyomatosis and RCC (HLRCC; AD)	FH	Papillary type 2, collecting duct; leiomyomas

Syndrome	Gene	GU tumour
Birt-Hogg-Dubé (AD)	FLCN	Hybrid oncocytic, chromophobe, oncocytoma
Hereditary breast–ovarian	BRCA2 > BRCA1	Prostate
Lynch	MMR genes	Upper-tract/bladder UC, prostate
Hereditary prostate cancer	HOXB13	Prostate → earlier PSA screening

KEY POINT The germline syndrome names the somatic driver (VHL → clear cell, MET → papillary, FH → type 2, BRCA → prostate) and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

11 · CLINICAL NGS AND BIOMARKERS

- Most commercial tests are targeted panels. **Tumour-only** panels cannot separate somatic from germline variants.
- **Variant allele frequency (VAF)** = fraction of reads with the variant; depends on zygosity, copy number, loss of heterozygosity, purity. Heterozygous variant × 60% purity → VAF 30%.

VAF	Interpretation
High	Clonal, early
Low	Subclonal, late
~50% / ~100%	Suspect heterozygous / homozygous **germline** variant
Single-digit % in cell-free DNA	Consider **clonal haematopoiesis**

- **Liquid biopsy**: cell-free DNA (cfDNA) contains a small circulating tumour DNA (ctDNA) fraction → deep sequencing; post-treatment ctDNA = minimal residual disease.

	Diagnostic	Prognostic	Predictive
Question	Is cancer present?	How aggressive?	Will this treatment help?
Prostate	PSA, urine PCA3, TMPRSS2–ERG	National Comprehensive Cancer Network (NCCN) risk groups, genomic classifiers	BRCA → PARP; germline HSD3B1

- Proving prediction needs a **biomarker-treatment interaction test** in a randomised trial; on interaction plots parallel lines = prognostic, crossing lines = predictive.
- Absolute benefit scales with baseline risk: 50% relative reduction gives 15% absolute benefit at 30% risk, only 1.5% at 3%.
- **HSD3B1 (1245A>C)** boosts dihydrotestosterone synthesis → ADT resistance but greater abiraterone sensitivity.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-gene-altered metastatic castration-resistant prostate cancer	Better radiographic progression-free and overall survival, driven by BRCA1/2	Germline + somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after immunotherapy	Longer overall survival	FGFR testing of all advanced UC

13 · MCQ ONE-LINERS

1. Feature unique to malignancy: local invasion.

2. Knudson two-hit: retinoblastoma; familial cases often bilateral.
3. Cyclin D–CDK4/6 phosphorylates RB1, releasing E2F.
4. INK4 inhibitors act only on CDK4/6.
5. MDM2 is the negative regulator of TP53.
6. HR needs sister chromatid — only S/G2.
7. ERCC2 mutation (NER) → cisplatin sensitivity in bladder cancer.
8. MMR defect raises mutation rate ~100-fold.
9. Extrinsic apoptosis is TP53-independent.
0. Telomeres lose ~100 bp per division.
1. GSTP1 hypermethylation: commonest somatic change in prostate cancer.
12. 5-methylcytosine deamination gives C→T transitions.
13. EZH2 writes H3K27me3.
14. siRNA silences one mRNA; miRNA silences many.
15. VHL is an E3 ubiquitin ligase targeting HIF-α.
16. Commonest prostate fusion: TMPRSS2–ERG.
17. SPOP: commonest mutated suppressor in primary prostate cancer.
18. PBRM1 mutation: better prognosis; BAP1: worse.
19. Largest consensus bladder subtype: basal/squamous.
20. Testis: 12p gain; TP53 mutation rare.
21. HLRCC gene: fumarate hydratase.
22. VAF ~50% without germline data: suspect germline variant.

55 Management of Abnormalities of the External Genitalia in Males

Reading time ≈ 66 min (slow pace)

1 · NORMAL DEVELOPMENT AND PENILE SIZE

- Male differentiation (weeks 9–13) needs testosterone (fetal Leydig cells) → dihydrotestosterone (DHT, via 5α-reductase) → functional androgen receptor. Any break → undervirilisation, not a new structure; female is the passive default.
- Genital tubercle → glans; urethral folds → shaft (fuse in midline over the urethral groove); labioscrotal swellings → scrotum (inferomedial migration + fusion = raphe). The glanular urethra probably forms by fusion of the urethral plate, not ectodermal ingrowth.
- **Stretched penile length (SPL)**: pubic symphysis (fat pad fully compressed) to glans tip; correlates with erect length better than flaccid length. Term neonate **3.5 ± 0.7 cm** (diameter 1.1 cm); adult 13.3 ± 1.6 cm.
- Micropenis cut-off (mean − 2.5 SD): **1.9 cm** at term–5 months; 1.5 cm at 30 weeks, 2.0 cm at 34 weeks; ≈ 3.7–3.9 cm at 5–11 years; adult 9.3 cm. Always compare with the cut-off column, not the mean.
- Minipuberty (first 3 months): loss of maternal oestrogen → gonadotrophin surge → second phase of penile growth.
- Tanner staging (1–5) of pubic hair, penis, testes: scrotal/testicular change first (stage 2), penile length stage 3, girth/glans stage 4. Used for varicocele (peaks Tanner 3) and timing of semen analysis (Tanner 5).

2 · PREPUCE: PHIMOSIS AND PARAPHIMOSIS

- **Physiological phimosis** at birth (glanular adhesions ± preputial ring). Smegma accumulation (first 3–4 years) and intermittent erections separate the prepuce: ≈ **90% retractable by 3 years**, < 1% phimotic at 17 years.
- **Secondary (pathological) phimosis**: scarring after forceful retraction, or balanitis xerotica obliterans (BXO). Forceful retraction is discouraged — it converts a physiological problem into a cicatricial one.
- Indications to improve retractility: persistent primary or secondary phimosis, balanitis, posthitis (preputial inflammation), BXO, urinary tract infection (UTI).
- **Topical steroid**: betamethasone 0.05% twice daily × 30 days (or thrice daily × 21 days) — equally effective; response within 2 weeks; ≈ 1 in 6 fails (more collagen, less elastic fibre in the prepuce).
- A retractile prepuce halves the risk of recurrent febrile UTI; steroid treatment of phimosis after a UTI (normal ultrasound [US]) reduces recurrence.
- Commonest acquired "penile lump": yellowish **entrapped smegma** under adhesions — released as adhesions resolve.
- **Paraphimosis** (retracted prepuce trapped behind the glans) — oedema builds within hours (faster the tighter the ring) and can progress to gangrene. Escalation: manual glans compression with distal traction (succeeds in most) → oedema reduction (iced glove 5 min, granulated sugar 1–2 h, multiple punctures) → dorsal slit → circumcision.

3 · CIRCUMCISION

Policy, technique, anaesthesia

- American Academy of Pediatrics (AAP) 2012: benefits (less UTI, heterosexual human immunodeficiency virus [HIV] acquisition, other sexually transmitted infections) insufficient to recommend routine neonatal circumcision, but sufficient to justify access and third-party payment.

- Devices: Gomco, Mogen, Plastibell — first separate the prepuce fully and inspect meatus/corona to exclude hypospadias or megameatus. Older children: freehand under general anaesthesia.
- Analgesia is mandatory: **dorsal penile nerve block superior to EMLA** (eutectic mixture of local anaesthetics — lidocaine/prilocaine; prilocaine → methaemoglobinaemia risk); ring block alternative.

Defer / avoid neonatal circumcision	Reason
Hypospadias, penile curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction; devices unsuitable
Large hydrocele / inguinal hernia	Push skin forward → secondary phimosis, buried or trapped penis
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
Neonatal intensive care unit setting	More complications

Benefits all follow from removing a moist, colonisable, potentially phimotic sac: penile cancer almost exclusively in men not circumcised at birth (phimosis a major risk factor); febrile infant UTI risk lower to at least 5 years (preputial colonisation by uropathogens), but **111** circumcisions prevent one UTI; less epididymitis and balanitis; ≈ **60%** protection against HIV acquisition in men (no reduction in transmission from infected men to partners); lower herpes simplex virus type 2 and high-risk human papillomavirus; no proven sexual dysfunction. Neonatal circumcision is more cost-effective than later.

Complications (overall 0.2–5%; rises 10–20-fold after infancy)

Complication	Management
Bleeding — commonest (≈ 0.1%), usually frenular; more in older children	Compression, ophthalmic cautery, silver nitrate, suture; occasionally haematoma evacuation
Wound infection (rare; Fournier gangrene reported)	Prevent with antibiotic ointment
Penile degloving (excess skin removed or edges not adherent)	Heals by re-epithelialisation: antibiotic ointment + warm baths; **no immediate suturing or graft**
Glanular adhesions / skin bridges (fat pad retraction, diaper irritation, incomplete lysis)	Adhesions separate spontaneously (rare after 9 y); office lysis with EMLA or vapocoolant; **steroids do not help**. Skin bridge (epithelialised; can cause chordee/torsion) → excise (office, or theatre if broad)
Inadequate/asymmetric excision	Revision needs general anaesthesia
Excess excision → chordee, torsion, lateral deviation	Skin flaps, Z-plasty, split-thickness graft
Trapped penis (cicatrix)	Betamethasone 0.05% + manual retraction → vertical relaxing incision → formal repair (flaps/Z-plasty)
Urethral injury	Urethroplasty matched to injury
Amputated glans	Re-attach within **8 h** (often without microscope)
Penile necrosis (cautery touching clamp/ring, lasers)	Phallic reconstruction vs female reassignment — latter problematic (male identity from androgen imprinting)

> **KEY POINT** Late sequelae are problems of skin quantity (too much → trapped penis, chordee, torsion; too little → redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis

- Almost exclusively in the **circumcised** penis (congenital form with hypospadias): meatitis from diaper irritation of the exposed meatus, inflammation after adhesion lysis, frenular devascularisation, BXO.

- Normal calibre (French, Fr; bougie à boule): < 4 months 6–10 Fr; 1–12 months 10–12 Fr; 1–10 years 12–14 Fr; > 10 years 14–18 Fr.
- Symptoms: upward-deflected stream (ventral web/baffle), thin forceful stream with straining/frequency/incomplete emptying, dysuria at start. Urethrography not needed unless UTI, incontinence or proximal anomaly suspected.
- Treatment: ventral meatotomy/meatoplasty long enough to restore calibre (office with EMLA, or general anaesthesia); suturing mucosa to glans with fine absorbable sutures reduces recurrence (sutureless clamp tailoring equally effective).

Balanitis xerotica obliterans (lichen sclerosus)

Chronic infiltrative cicatrising dermatosis → pathological phimosis; may involve glans, meatus, urethra. Presents with non-retractile prepuce at puberty (rare < 5 years), irritation, bleeding, post-void discomfort, occasionally retention. Cause unknown. Steroids only for mild preputial disease; **circumcision is preferred** ± meatotomy/meatoplasty; ≈ 1/5 with meatal disease need further meatal surgery — follow them; buccal mucosa urethroplasty for severe urethral disease. Prepuce-sparing option: triradiate preputioplasty + intralesional triamcinolone (less meatal stenosis).

4 · INCONSPICUOUS PENIS VERSUS MICROPENIS

> **KEY POINT** **Stretched length is the single discriminator.** Inconspicuous penis (buried, webbed, trapped) = normal SPL and diameter hidden by skin, fat or scar → anatomical treatment. Micropenis = small but normally formed organ → endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital (poor penopubic/penoscrotal fixation)	Inelastic dartos; skin not anchored to deep fascia, restricting extension	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss first; liposuction only in adolescents (prepubertal fat pad shrinks with growth)
Trapped (after circumcision)	Cicatrix embeds penis in fat	Office dilation with fine haemostat or betamethasone + retraction → repair under general anaesthesia at ≥ 6 months

- Examination: retract skin lateral to the shaft — if the glans exposes easily the shaft is normal; distinguish from a transient large infant fat pad, which resolves with walking.
- Trapped penis follows circumcision with large hernia/hydrocele, of a webbed penis, or of a shaft retracting into the scrotum. Risks: difficult voiding, urine pooling → balanitis, UTI; psychosocial.
- **Repair sequence**: excise dysgenic dartos bands (mainly proximal dorsal shaft) → unfurl prepuce for ventral cover → fix dorsal subcutaneous tissue to pubic fascia → fix scrotal subcutaneous tissue to ventral penile base (spare neurovascular bundles). Skin options: penoscrotal or lateral Z-plasty, ventral preputial island flap, graft. Rarely divide suspensory ligament and excise suprapubic fat (protect cords).
- **Webbed penis** (penoscrotal fusion): scrotal skin on ventral shaft; congenital (normal penis/urethra) or after excess ventral skin removal. Repair: scrotal fixation as above; transverse incision of web + vertical closure; or circumferential incision 1.5 cm below corona with Byars' preputial flap to ventrum. Avoid neonatal circumcision. Rarely hypoplastic distal urethra needs reconstruction.

Micropenis

- Normally formed penis ≥ 2.5 SD below mean SPL (< **1.9 cm** at term); length-to-girth ratio normal. Testes often small/cryptorchid; scrotum fused, small.
- Embryology: differentiation complete by week 12 under placental human chorionic gonadotrophin (hCG)-driven testosterone (defect → small penis *with hypospadias*); later growth needs fetal

luteinising hormone (LH) → gonadotrophin deficiency after 14 weeks gives a normally formed micropenis.

- Chromosomal associations: Klinefelter (47,XXY), other X polysomies, trisomies 8, 13, 18.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism (hypoglycaemia, growth hormone deficiency), midline brain defects	Low LH/FSH (follicle-stimulating hormone)
Hypergonadotrophic (primary testicular failure)	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

History, examination, **karyotype**, SPL, corpora, cryptorchidism → paediatric endocrinologist → testosterone before/after hCG stimulation (± gonadotrophin-releasing hormone [GnRH] test) → anterior pituitary screen (serial glucose, Na/K, cortisol, thyroid) → magnetic resonance imaging (MRI) brain (hypothalamus, pituitary, midline) → **test end-organ response: intramuscular testosterone enanthate for 3 months** (short courses do not affect height; prolonged courses advance bone age); repeat at puberty.

Outcome: male identity, voiding standing and satisfactory sexual function in most, though some are dissatisfied with size.

5 · ABNORMAL ORIENTATION: CURVATURE (CHORDEE WITHOUT HYPOSPADIAS) AND TORSION

Curvature may be congenital or acquired (circumcision, hypospadias repair, trauma). Repair when sexual dysfunction is expected, with concurrent penile surgery, or when growth is likely to worsen it.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; isolated chordee with dorsal hood/deficient ventral skin; prominent frenulum → distal chordee	Catheter in urethra → deglove → excise fibrous tissue superficial to Buck's fascia → Byars' flaps → persistent: dorsal plication/Nesbit/corporal rotation → short urethra: urethral reconstruction; frenular: frenulotomy
Dorsal	Isolated, epispadias	Deglove → release → ventral plication/ellipse; protect dorsal neurovascular bundles
Lateral	Corporal over/undergrowth, asymmetric skin excision; straight when flaccid — noticed only on erection, later childhood	Plication/ellipse of the corpus **opposite** the maximal curvature

Principle: superficial → deep; release skin and tethers first, plicate the convex side only if curvature persists. **Artificial erection** (saline into a corpus ± tourniquet) locates the apex and confirms correction. → *Paediatric Urology book, Ch 56* for hypospadias.

Penile torsion: usually **counter-clockwise (to the left)**, < 60°; raphe spirals; anomalous skin arrangement; corpora at the base normally oriented — a skin-level deformity. May accompany hypospadias or hood. Avoid neonatal circumcision if repair planned. < 90°: deglove, rotate glans opposite (clockwise), suture to skin — serpentine raphe acceptable if meatus normal. ≥ 90°: mobilise base, divide dysgenic bands, non-absorbable anchoring sutures at corporal base ± dorsal dartos flap or urethral mobilisation.

6 · ABNORMAL NUMBER, MASSES AND LYMPHOEDEMA

Feature	Aphallia (penile agenesis)	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, scrotum + testes normal	1 in 5 million; accessory penis to complete duplication
Associations	Cryptorchidism, vesicoureteral reflux (VUR), renal agenesis, horseshoe kidney, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI → neophallus (potentially fertile male) vs reassignment (multidisciplinary; reassigned children may develop male identity)	Renal US + voiding cystourethrogram (VCUG) + penile US/MRI → individualised

Aphallia — urethral opening	Frequency	Survival / anomalies
Postsphincteric (anal verge, skin tag)	Commonest	Best; fewest anomalies
Presphincteric (prostatorectal fistula)	Intermediate	Intermediate
Urethral atresia + vesicorectal fistula	Least	Fatal; most anomalies

- Penile masses: cysts commonest — parameatal (excise + marsupialise, avoid meatal stenosis), median raphe (excise if symptomatic), epidermal inclusion after surgery (excise); naevus (excise for diagnosis); **juvenile xanthogranuloma — observe 1 year** (self-limiting).
- Genital lymphoedema: congenital (Milroy — autosomal dominant, at birth; Meige — 1st–2nd decade; praecox — puberty, ≈ 80%) or acquired (anogenital Crohn disease — medical therapy incl. anti-tumour necrosis factor agents). Associated with Turner, Noonan, Klinefelter. Surgery is excisional, not drainage: all tissue between Buck's fascia and skin; scrotum excised except posterior skin (preserve cords/testes); cover with local flaps or grafts; counsel recurrence.
- Vascular: haemangiomas involute (observe; propranolol/steroid/laser if ulcerated); malformations persist or enlarge ("bag of worms" not decompressing supine) → Doppler/computed tomography (CT)/MRI → en bloc resection ± preoperative embolisation. Klippel–Trénaunay–Weber: genital lesions bleed.

7 · URETHRAL ANOMALIES

- **Congenital urethral fistula** (coronal/subcoronal, normal meatus; possibly focal urethral-plate defect): circumscribe + multilayer closure, or open ventral glans and Thiersch–Duplay if glans bridge thin.
- **Urethral duplication**: usually sagittal (≈ 94%); the **ventral channel is the functional urethra** (sphincter, verumontanum). **Effmann**: type I incomplete (blind; IA distal, IB proximal), II complete (IIA two meatuses — IIA2 includes "Y-type" perineal; IIB one meatus), III with bladder duplication. Presentation: double stream, incontinence (more proximal accessory origin → more incontinence), UTI, chordee, obstruction from a bifurcation flap. VUR commonest association; midline defects. Evaluate with VCUG, retrograde urethrogram, cystoscopy. Treat if symptomatic; **never use the hypoplastic dorsal accessory urethra as the main channel** — excise, fulgurate/sclerose, septotomy or urethro-urethrostomy.
- † **Megameatus intact prepuce (MIP)**: hypospadias variant with a normal, complete prepuce hiding a wide glanular/coronal meatus with a deep glanular cleft and no chordee; often discovered during or after circumcision (stop and preserve the prepuce if recognised). Repair is cosmetic/functional: pyramid procedure or glans approximation/tubularisation.
- † **Anterior urethral valves / diverticulum**: far rarer than posterior valves; the "valve" is often the distal lip of a ventral diverticulum; commonest in the bulbar urethra, then penoscrotal and penile. Presents with poor stream, UTI, post-void dribbling, ventral penile swelling on voiding; may cause bladder and upper-tract damage. Diagnosis on VCUG. Treatment: endoscopic valve ablation; large diverticulum → open excision with urethroplasty.

- † **Cowper syringocele** — cystic dilatation of the bulbourethral (Cowper) gland duct. Maizels types: simple (minimal dilatation), perforated (drains widely into urethra, diverticulum-like), imperforate (submucosal cyst, obstructive), ruptured (membrane remnant). Symptomatic (obstruction, UTI, haematuria, dribbling) → endoscopic unroofing/marsupialisation.
- † **Megalourethra** — dilated penile urethra from deficient erectile tissue. **Scaphoid** (commoner): corpus spongiosum deficient, ventral ballooning on voiding, cavernosa normal. **Fusiform** (rarer, severe): spongiosum and cavernosa deficient, circumferential dilatation, flaccid penis, often lethal associated anomalies. Both associated with prune-belly syndrome and upper-tract anomalies; repair by reduction (tailoring) urethroplasty.
- Meatal stenosis — section 3.

8 · PRIAPISM

Erection > 4 hours without stimulation; usually corpora cavernosa only. **Corporal blood gas** decides urgency: ischaemic priapism is a compartment emergency; arterial priapism is a well-oxygenated fistula that can be observed.

Type	Cause in children	Corporal gas / exam	Treatment
Ischaemic (low-flow)	**Sickle cell disease** (commonest), leukaemia, other haemoglobinopathies	Dark, hypoxic, acidotic; rigid, painful	Hydration, oxygen, alkalinisation, analgesia ± transfusion → aspiration/irrigation ± intracavernous α-agonist (phenylephrine) → shunt
Non-ischaemic (high-flow)	Straddle injury → arteriocavernous fistula (colour Doppler)	Bright red; not rigid or painful	Observe → superselective embolisation
Stuttering	Sickle cell	Short recurrent episodes	Bedtime oral pseudoephedrine; oral β-agonist; long-term phosphodiesterase-5 inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	—	None — resolves in 2–6 days

- Sickle mechanism: sleep hypoventilation lowers corporal pH/oxygen → sinusoidal sickling → stasis → self-perpetuating cycle.
- Shunts: distal cavernoglanular (Winter, Ebbehoj, Al-Ghorab) → proximal cavernospongiosal (Quackels, Sacher) → saphenous–cavernous (Grayhack). Counsel about fibrosis and erectile dysfunction.

9 · SCROTAL ANOMALIES

- **Penoscrotal transposition**: failed inferomedial migration of labioscrotal swellings; partial forms — shawl, doughnut, prepenile scrotum. Associated with perineal/scrotal hypospadias + chordee, caudal regression, sex chromosome anomalies, Aarskog–Scott; with complete transposition, **up to 75%** have significant upper-tract anomalies → renal US + VCUG.
- Repair (usually with hypospadias repair) aims to free penis, relocate scrotum dependently: **rotation-flap scrotoplasty** — circumscribe the upper border of each hemiscrotum (≥ half of scrotum), join medial ends on ventral penis and down the raphe → lidocaine–epinephrine infiltration → deepen sparing tunica vaginalis and cord → rotate wings medially under the penis and suture in midline. Alternative: buttonhole penile relocation through a suprapubic skin/fat plug.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds, no raphe	Proximal hypospadias	Scrotoplasty/Z-plasty with hypospadias repair

Anomaly	Feature	Association	Management
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal; gubernacular defect	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism, hernia; perineal lipoma usually means no other anomaly	Upper-tract US; scrotoplasty + orchidopexy at 6–12 months
Scrotal hypoplasia / agenesis	Underdeveloped / absent	UDT; agenesis — all XY, bilateral UDT	—
Scrotoschisis	Scrotal rupture, extruded testis	Meconium peritonitis	Exploration, orchidopexy, closure

10 · INGUINAL HERNIA AND HYDROCELE

All are variations on where and how completely the **processus vaginalis** (formed in month 3, obliterates after descent) failed to close; only ≈ 18% are closed at term.

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; abdominal contents enter	Repair soon after diagnosis
Communicating hydrocele	Patent to testis, fluid only	Repair if persists > 1–2 y, enlarges or fluctuates
Hydrocele of cord (encysted)	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; extends through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia in 1–5% of children; prematurity (≈ 13% < 32 weeks, ≈ 30% if < 1 kg); right 3:1 (later right descent). Bilaterality risk: female, premature, < 1 year, cryptorchidism. Ascites/ventriculoperitoneal shunt raise risk.
- **Intermittent bulge** = patent processus → surgery; fixed collection → observe. Provoke by crying, coughing, blowing balloons; parent photographs are reliable; silk-stocking sign. Cord hydrocele: normal cord felt above and below. Neonatal bowel also transilluminates. US only for obesity, incarceration or doubt.
- Incarceration: irritable child, irreducible bulge, vomiting, distension → immediate exploration.
- Timing: reducible → elective within weeks; difficult reduction → urgent. Preterm: repair before discharge, or delay in very sick/< 1500 g infants (some wait to 50–52 weeks post-conception).

ALGORITHM
Skin-crease incision → Scarpa's fascia → open external oblique along fibres sparing ilioinguinal nerve (in young infants rings coincide — Mitchell-Banks, external oblique not opened) → split cremaster → find sac anteromedially → separate atraumatically from vas and vessels, divide → dissect to internal ring and **high-ligate** (open sac first if contents suspected) → tighten a wide ring medially → excise redundant tunica, remove appendages → **confirm testis is scrotal**.

- Peripubertal non-communicating hydrocele (> 10–12 y) → scrotal approach; convert to inguinal if communication found.
- Recurrence 0.5–1% (2% preterm, 3–6% after incarceration), mostly within 2 years: missed/torn sac, damaged floor, missed direct hernia → inguinal re-repair.
- Iatrogenic cryptorchidism (orchidopexy at herniorrhaphy if testis high/retractile); atrophy (incarceration in infants); vas injury (may cause later obstruction). Persistent hydrocele after repair → observe 1 year → scrotal repair.
- Scrotal approach (ligation at external ring) — early data only. Laparoscopic repair (intraperitoneal or extraperitoneal closure) now matches open; advantages: sees and closes contralateral ring,

reduces incarcerated bowel under vision; open remains surgeons' preference for lowest recurrence.

- **Abdominoscrotal hydrocele**: closed-processus hydrocele expanding upward under pressure (exudate, lymphatic obstruction); may cause hydroureteronephrosis or limb oedema; testis elongation usually reversible. Inguinal excision (aspirate scrotal part first; leave a strip on the cord to protect vas) ± orchidopexy; midline/laparoscopic for large bilateral; primary scrotal drainage-plication is an option.
- Contralateral patency ≈ 40% infants → 20% after 10 years, but metachronous hernia only ≈ **7%** — argument against routine contralateral exploration; laparoscopy through the sac sees the other ring (> 99% accurate).

11 · ACUTE SCROTUM

Only cord torsion is time-critical; the clinical and imaging strategy exists to rule it in or out quickly. Differential also includes trauma, hernia, orchitis (mumps, Henoch–Schönlein purpura), tumour, and **referred pain** (ureteric calculus, appendicitis).

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; nausea/vomiting common	Sudden or insidious; worse with activity; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present
Signs	High-riding, horizontal lie, diffuse tenderness	**Blue dot**, tender upper-pole nodule	Enlarged tender epididymis; pyuria 20–40%
Colour Doppler US (CDUS)	Absent/reduced flow, heterogeneity; **whirlpool/snail sign**	Avascular upper-pole lesion, epididymal hyperaemia	Increased epididymal flow
Management	**Emergency exploration**	Ice, non-steroidal anti-inflammatory drugs (NSAIDs), rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- **Bell-clapper** deformity (high tunica attachment, ≈ 12% at autopsy); familial cases; cryptorchid testes at higher risk; torsion after orchidopexy from suture failure. Left-sided; incidence ≈ 1 in 4000.
- Delay to presentation (embarrassment, low awareness) is the main cause of loss — educate boys and parents.
- Cremasteric reflex (present normally after 2 years) may be present in up to 10% of proven torsions — presence does not exclude torsion. Landmarks are lost as duration increases.
- Predictors: absent reflex, nausea/vomiting, scrotal skin changes. **TWIST score** (validated) stratifies risk: † testis swelling 2, hard testis 2, absent cremasteric reflex 1, nausea/vomiting 1, high-riding testis 1; ≥ 5 high risk → theatre without imaging; ≤ 2 low risk.
- CDUS: absent flow is highly specific, but sensitivity only 63–90% (residual flow in partial torsion; operator-dependent). High-resolution US of the whole cord: twist seen in almost all torsions; a linear cord argues against. Heterogeneous echotexture = necrosis. Urinalysis helps only with epididymitis/calculus. **Do not delay surgery for imaging when suspicion is high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	**≈ 5%**	20%	40%	60%	80%	90%

Degree of twist also matters (lost testes ≈ 540° vs 360°); in-hospital delay and interhospital transfer lower salvage.

ALGORITHM
Suspected torsion → scrotal (hemiscrotal or raphe) exploration → affected side first: open tunica, note colour, twists, anatomy → detorse, warm moist packs → **meanwhile fix contralateral testis with non-absorbable suture** → reassess (subjective; Doppler or tunica albuginea incision to see bleeding unvalidated): viable → dartos pouch or direct dartos fixation (non-absorbable); non-viable → orchidectomy (cord ligated in segments); prosthesis after healing or in puberty.

- Manual detorsion fails in ≈ 1/3 (twist opposite to that assumed); success still needs orchidopexy.
- Compartment syndrome: tunica albuginea incision ± tunica vaginalis flap lowers pressure — unproven.
- Late: atrophy even after salvage (predicted by > 1 day pain, heterogeneity); subtle semen defects and low inhibin B from ischaemia–reperfusion (not antisperm antibodies).
- **Intermittent torsion**: precedes acute torsion in 30–50%; abrupt self-limiting pain; exam/CDUS often normal between episodes → **elective bilateral orchidopexy** (counsel that proof may be impossible).

Perinatal (extravaginal) torsion

Feature	Intravaginal	Extravaginal (perinatal)
Anatomy	Bell-clapper; twist within tunica	Whole cord + tunica twist before scrotal fixation
Presentation	Acute pain, high horizontal testis	Hard painless testis, discoloured fixed skin, or vanished testis/nubbin
Salvage	Time-dependent	Rare; bilateral 5–22%
Urgency	Emergency	Debated; urgent if after a documented normal postnatal exam

Immediate exploration offers possible salvage and detects contralateral torsion; elective exploration avoids neonatal anaesthesia for a rarely salvageable testis. Inguinal approach ligates a patent processus and allows tumour exclusion.

Appendage torsion, epididymitis, others

- Appendix testis (Müllerian, ≈ 80% of testes) and appendix epididymis (Wolffian); pedunculated type twists. CDUS: round avascular upper-pole lesion with surrounding hyperaemia (hypoechoic within 24 h). Blue dot can be falsely positive in cord torsion. Explore only if cord torsion cannot be excluded or pain prolonged/recurrent.
- Epididymitis: ask about UTI, sexual activity, catheterisation, dysfunctional voiding. Ice, NSAIDs, elevation; broad-spectrum antibiotics (Gram-negative) if pyuria; consider sexually transmitted infection treatment in adolescents. Prepubertal boy with positive urine culture → renal US + VCUG (ectopic ureter, urethral or ejaculatory duct anomaly); recurrent → endoscopy; utricle/ejaculatory duct cyst excision ± vasectomy.
- Idiopathic scrotal oedema (prepubertal, eosinophilia, **fountain sign**; self-limiting, antihistamine); Henoch–Schönlein purpura (may cause torsion — explore if suggested); cystic dysplasia of rete testis (ipsilateral renal agenesis/dysplasia; enucleation or observation); testicular rupture (CDUS; surgery or selective conservative).

12 · ADOLESCENT VARICOCELE

- Prevalence ≈ **15%** (as in adults); appears after 10 years, peaks **Tanner 3**; tall thin boys; familial (brothers).
- Left-sided because of high left renal vein pressure, incompetent valves at the left spermatic–renal junction, collaterals, nutcracker (left renal vein between aorta and superior mesenteric artery). Subclinical right reflux common.

- **Isolated right or non-decompressing varicocele** → abdominal/pelvic imaging for a mass.
- Examine supine and standing, at rest and Valsalva. Grading: 0 subclinical (CDUS only); 1 palpable with Valsalva; 2 palpable at rest; 3 visible.
- Testicular volume by **US (L × W × H × 0.71)** — most sensitive for asymmetry (Prader/Takihara orchidometers acceptable); volume doubles Tanner 2→3, so serial measures. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Presumed thermal injury. Surrogates (hypotrophy 10–20% or 2–3 mL, grade, hormones, histology) do not reliably predict adult semen; catch-up growth after repair may also occur spontaneously. Semen analysis is the best indicator once growth stabilises. Hence **observation is the default**.
- Intratesticular varicocele (veins > 2 mm near mediastinum, Valsalva flow) with asymmetry → repair.

> KEY POINT **Indications for repair**: left testis > **20%** smaller (or bilateral hypotrophy), pain (rare; relieved in most), abnormal semen analysis (reliable at Tanner 5 / ≥ 18 y), intratesticular varicocele with asymmetry. Total testicular volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Simple; hydrocele = lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy; microscope > loupes
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence; maps collaterals

- **Open Palomo**: muscle-splitting incision medial to anterior superior iliac spine → peritoneum medially → isolate and ligate spermatic vessel bundle.
- **Laparoscopic**: three ports → open peritoneum above internal ring → identify artery (Doppler/papaverine) and lymphatics (magnification; paratesticular dye risks intratesticular injection) → clip/divide veins, including veins accompanying the artery (prevents recurrence).
- Atrophy risk: artery ligation followed by later surgery damaging vasal vessels (hernia repair, vasectomy).
- Mass ligation trades hydrocele for certainty; sparing trades recurrence for lymphatics. Hydroceles may appear ≥ 2 years later; up to half need surgery/aspiration. Genitofemoral nerve injury (cautery) → transient thigh paraesthesia.
- Sclerotherapy: retrograde transfemoral venography (technical failure 5–22%; radiation) or antegrade via scrotal vein cannulation; agents sodium tetradecyl sulfate, polidocanol, coils.

13 · EPIDIDYMAL AND VASAL ANOMALIES

- Epididymal cysts: common, rise with age; spermatoceles are postpubertal and contain sperm; distinct from von Hippel–Lindau cystadenoma; mostly observe.
- Congenital absence of vas: bilateral → **CFTR** (cystic fibrosis transmembrane conductance regulator) mutations; unilateral → **renal anomalies** (Wolffian origin).
- Persistent mesonephric duct: vasoureteral fusion with ipsilateral renal and seminal vesicle agenesis → reflux into Wolffian remnants → UTI, epididymo-orchitis.
- Testicular microlithiasis → *Paediatric Urology book, Ch 57*; undescended testis → *Paediatric Urology book, Ch 57*.

14 · MCQ ONE-LINERS

1. Micropenis in a term neonate: stretched length < 1.9 cm (−2.5 SD).
2. Commonest cause of micropenis: hypogonadotrophic hypogonadism.
3. First therapeutic step in micropenis: intramuscular testosterone enanthate for 3 months to test response.
4. Buried penis has normal stretched length — distinguishes it from micropenis.
5. Phimosis topical therapy: betamethasone 0.05%; ≈ 1 in 6 fail.
6. Commonest circumcision complication: bleeding, usually frenular.
7. Circumcision degloving: heal by secondary intention, no immediate graft.
8. Meatal stenosis occurs almost exclusively after circumcision.
9. Best neonatal circumcision analgesia: dorsal penile nerve block (better than EMLA).
0. Amputated glans: reattach within 8 hours.
1. Penile torsion is usually counter-clockwise; corpora at base normal.
2. Lateral curvature: plicate the corpus opposite the maximal curvature.
3. Urethral duplication: ventral urethra is functional; Effmann classification.
4. Megameatus intact prepuce: hypospadias variant hidden by a normal foreskin.
5. Fusiform megalourethra: spongiosum and cavernosa deficient; prune-belly association.
6. Anterior urethral valve: commonest bulbar; often distal lip of a diverticulum.
7. Aphallia: commonest type has postsphincteric opening at anal verge, best survival.
18. Complete penoscrotal transposition: up to 75% have upper-tract anomalies.
19. Suprainguinal ectopic scrotum: ipsilateral renal agenesis — image upper tracts.
20. Commonest cause of paediatric priapism: sickle cell disease (ischaemic).
21. High-flow priapism follows straddle injury; embolise if persistent.
22. Neonatal priapism: no treatment, resolves in 2–6 days.
23. Commonest acute scrotum cause overall (prepubertal): torsion of appendix testis.
24. Present cremasteric reflex does not exclude testicular torsion.
25. Whirlpool/snail sign on high-resolution US = twisted spermatic cord.
26. Torsion < 6 hours: orchidectomy ≈ 5%; > 24 hours ≈ 80%.
27. Always fix the contralateral testis with non-absorbable suture in torsion.
28. Perinatal torsion is extravaginal; bilateral in 5–22%.
29. Paediatric hernia recurrence after open repair: 0.5–1%.
30. Abdominoscrotal hydrocele: processus closed; no ring tightening needed.
31. Varicocele peaks at Tanner 3; repair if > 20% volume discrepancy.
32. Lowest hydrocele + recurrence: microscopic subinguinal varicocelectomy.
33. Isolated right varicocele → image abdomen for mass.
34. Bilateral absent vas = CFTR mutation; unilateral absent vas = renal anomaly.

58 Management of Abnormalities of the Genitalia in Girls

Reading time ≈ 22 min (slow pace)

1 · EMBRYOLOGY THAT EXPLAINS THE ANOMALIES

- **Cloaca** (hindgut + allantois) is divided by the **urorectal septum** (Tourneux fold + Rathke plicae) into urogenital sinus (ventral) and anorectal canal (dorsal); fuses with cloacal membrane by weeks 6–7. The node = **perineal body**; it splits the cloacal sphincter into urogenital diaphragm and external anal sphincter → **both pudendal-innervated**.
- Müllerian ducts arise lateral to the mesonephric (wolffian) ducts → **müllerian anomaly + ipsilateral renal anomaly**. Fused ducts → uterus, cervix, upper two-thirds of vagina.
- Müllerian tubercle induces **sinovaginal bulbs** → vaginal plate → canalised caudal-to-cranial → distal vagina. Distal sinus everts → vestibule; hymen = posterior sinus-wall invagination.
- Remnants: Skene glands (≈ prostate), Bartholin (≈ bulbourethral), Gartner duct (wolffian).
- **Caudal regression** (Duhamel): caudal somite insult at weeks 4–5 (<10 mm) → bowel, kidney, bladder, müllerian and lumbosacral defects; maternal diabetes. Earlier insult → higher anorectal malformation and worse associated anomalies. Cranial extension → VATER (vertebral, anal, tracheo-oesophageal, radial/renal) and MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · EVALUATION AND CLASSIFICATION

- Frog-leg position; record clitoral size, hymen, meatal site, character of any mass. **Pull-down manoeuvre** (labia majora drawn caudally and laterally) displays the distal vagina. Probe/feeding tube defines meatus. Discharge or bleeding → vaginoscopy under anaesthesia.
- Ultrasonography (US) for anatomy and upper tracts; **magnetic resonance imaging (MRI) = gold standard** for müllerian anatomy (cervix? functioning endometrium?). Broadest classification: American Society for Reproductive Medicine (ASRM).

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	Mayer–Rokitansky–Küster–Hauser (MRKH) syndrome
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, Herlyn–Werner–Wunderlich syndrome (HWWS)

> **KEY POINT** Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · CLITORIS

- **Clitoromegaly**: commonest cause **congenital adrenal hyperplasia (CAH)** → electrolytes, 17-hydroxyprogesterone, karyotype in every such newborn; steroid replacement stops further growth → *Ch 59*. Others: maternal androgen tumour, Turner mosaic disorder of sex development (DSD) with testicular tissue, plexiform neurofibroma. Palpate corpora: normal corpora + large hood = **pseudo-enlargement** (lymphatic).
- **Diminutive clitoris** → complete androgen insensitivity syndrome (CAIS).

4 · VESTIBULE: THE INTERLABIAL MASS

- **Urethral prolapse**: circumferential mucosal eversion; prepubertal African-ancestry girls and postmenopausal women. Spotting; **doughnut with central meatus** — catheter through centre →

no imaging. Sitz baths, topical steroid/oestrogen (recurrence high) → **circumferential excision** with mucosa-to-vestibule suture; avoid ligation over catheter and cryosurgery.

- **Urethral polyp**: paediatric caruncle; benign; excise.
- **Paraurethral (Skene) cyst**: maternal oestrogen; **displaced meatus, eccentric stream**; usually ruptures; needle drainage if persistent.
- **Gartner cyst / covered ectopic ureter**: anteromedial vaginal wall; an ectopic ureter ending in a wolffian remnant. Contrast defines anatomy; incise. Upper-pole heminephrectomy if wetting or infection persists.
- **Prolapsed ureterocele**: infant, retention, pink → dusky; duplex upper-pole hydronephrosis. Decompress (needle/incision) → reduce → catheter, usually under general anaesthesia.
- **Ectopic ureter** distal to sphincter → **daytime wetting despite normal voiding** († classically continuous dribbling — the girl who is never dry); single-system ectopia may mimic absent kidney.

> **KEY POINT** Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Urethral- or vaginal-origin cyst → **always renal US**.

- **Labial swelling**: commonest cause inguinal hernia (see §8). **Childhood asymmetric labium majus enlargement (CALME)**: unencapsulated physiological tissue → observe; growth or a defined mass → exclude neurofibroma, sarcoma, vascular malformation, Crohn disease.

5 · VAGINA: OBSTRUCTION AND ABSENCE

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate mucocolpos); bluish + amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through ± flaps
MRKH	Bulbs/plate; absent	Hymenal fringe + pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy + neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

Imperforate hymen

Probably the commonest obstructive anomaly. **Transverse** incision avoids extension into urethra or rectum; aspiration alone leaves residue → ascending infection. Hymenal injury (laceration, transection) suggests abuse → anaesthesia examination and forensic swabs.

Transverse septum

- ~1 in 70,000; amenorrhoea, pain, retention; often <1 cm with small perforation that still obstructs. Delay → retrograde menstruation → endometriosis.
- **MRI distinguishes high septum from cervical agenesis** — misdiagnosis is disastrous.
- If pain controlled, gonadotropin-releasing hormone (GnRH) agonist or continuous pill lets the lower segment dilate, easing repair.
- Options: incision; excision with mucosal approximation; **Z-plasty** (converts a contracting transverse scar to longitudinal); balloon division (US-guided angiocatheter → guidewire → balloon → excise); transabdominal approach where an intact hymen matters.
- Main complication: **stenosis** → Z-plasty, mould. **Hollow mould** when a uterus exists (menses), solid after neovagina.

Distal vaginal atresia

Sinus fails; uterus and upper vagina normal. US/MRI mandatory. Transverse incision → dissect to distended vagina (haematocolpos acts as tissue expander) → drain → pull-through; flaps or abdominoperineal approach for gaps.

MRKH (müllerian agenesis)

- ~1 in 5000; 46,XX, normal gonadotropins, **functioning ovaries**; second commonest cause of primary amenorrhoea after gonadal dysgenesis. Some have a functioning rudimentary uterus (cyclic pain).
- Renal anomaly ≈ **one-third** (agenesis/ectopia); skeletal 10–20%; aberrant pelvic vessels. No lumbosacral dysraphism.

	Type A (typical)	Type B (atypical)
Uterine buds / tubes	Symmetrical / normal	Asymmetrical / abnormal
Renal, Klippel–Feil, MURCS	Absent	Present → full renal and skeletal work-up

DSD causes of absent vagina (CAIS) → *Ch 59*; urogenital sinus and cloacal reconstruction → *Ch 60*.

6 · VAGINAL REPLACEMENT

- **Dilation first-line** (Frank; Ingram bicycle seat): needs a dimple and compliance; surgery not superior.
- **Vecchietti**: olive on dimple, traction sutures through abdomen (laparoscopic).
- **McIndoe (Abbe)**: bowel preparation → split-thickness buttock graft over stent → dissect urethrorectal space to peritoneum → labia sutured over stent, suprapubic catheter, bed rest 1 week. Stenosis → dilation required. Full-thickness graft or labial flaps (Williams pouch) contract less.
- **Buccal mucosa**: non-keratinised, well-vascularised; also for revisions. **Robotic tubularised peritoneal vaginoplasty**: no bowel or hair removal.
- **Sigmoid** (default bowel): distal pedicled segment, ~3 cm intervening bowel excised (no overlapping suture lines → fewer fistulae), fixed to posterior peritoneum (prevents prolapse). Intact cervix → anastomose for menses. **Ileum** stenoses more; reserved for irradiated pelvis or absent colon. Mucosal prolapse → trim; stenosis → Z-plasty.

Bowel	Skin (McIndoe)
Lubricates, less contracture	Stenosis → mandatory dilation
Discharge; adenocarcinoma	Squamous cell carcinoma, condylomata

All neovaginas → **annual examination**. Timing by diagnosis: skin needs maturity for dilation.

> KEY POINT **Cervical atresia**: perineum-to-uterus anastomosis loses the endocervical barrier → life-threatening ascending infection. Hysterectomy + neovagina is often best; otherwise endometrial suppression and in vitro fertilisation with transmyometrial transfer.

7 · LATERAL FUSION ANOMALIES

- **Didelphys**: failed medial-wall resorption; septate vagina up to 75%; excise septum only for dyspareunia, delivery obstruction or two tampons. **Septate uterus**: incise only for recurrent pregnancy loss; bicornuate = incomplete fusion.
- **HWWS**: didelphys + unilateral imperforate hemivagina → pain **without amenorrhoea**; unilateral mass, lateral bluish bulge; **ipsilateral renal agenesis**. Wide septal incision prevents endometriosis. Zhu class 1 complete (earlier), class 2 perforated (commoner, later).

8 · ACQUIRED DISORDERS

- **Vaginal rhabdomyosarcoma**: grapelike bleeding mass, <2 years; best prognosis of genital primaries (embryonal). Biopsy → computed tomography (CT) chest/abdomen + marrow → chemotherapy ± local resection. Intergroup Rhabdomyosarcoma Study (IRS) trials cut surgery from **100% to 13%**.
- **Labial adhesions**: commonest interlabial abnormality; peak 13–23 months; minora fuse from fourchette forward (majora fusion → DSD). Post-void dribbling. **80% resolve in 1 year** → treat only symptoms: topical oestrogen (breast budding, pigmentation) or betamethasone 0.05% with daily pull-down → office separation under eutectic mixture of local anaesthetics (EMLA) → surgical lysis. Recurrence depends on **family compliance**, not method.
- **Female genital mutilation (FGM)**: ~200 million; typically 4–10 years. **World Health Organization (WHO)**: I clitoridectomy, II excision, **III infibulation**, IV other. Sequelae: pain, infections, retained menses, vaginal calculus, apareunia. Midline scar division with edge closure succeeds; clitoral reconstruction less predictable.
- **Condylomata** (human papillomavirus, HPV): suspect abuse, but perinatal transmission occurs; >50% resolve; imiquimod, CO_2 laser; vaccination prevents.
- **Inguinal hernia**: may be CAIS presentation (~1–2.5%) → uterus on US, vaginoscopic cervix, or karyotype; laparoscopy: **round ligament absent in genetic males**. If CAIS, preserve testes (aromatised oestrogen) → *Ch 59*.

9 · MCQ ONE-LINERS

1. Urogenital diaphragm and external anal sphincter share pudendal innervation.
2. Müllerian anomalies travel with ipsilateral renal anomalies.
3. Caudal regression: weeks 4–5 insult; maternal diabetes.
4. MURCS: müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia.
5. MRI is gold standard for müllerian anatomy.
6. Commonest cause of clitoromegaly: CAH.
7. Diminutive clitoris suggests CAIS.
8. Urethral prolapse: doughnut with central meatus; excise circumferentially.
9. Skene cyst: displaced meatus, eccentric stream.
0. Ectopic ureter below sphincter: daytime wetting despite normal voiding.
1. Any introital cyst: image the kidneys.
2. Imperforate hymen: transverse incision, never simple aspiration.
13. Commonest transverse septum level: upper/middle third.
14. Commonest septum-repair complication: stenosis.
15. Hollow mould when uterus present.
16. MRKH: second commonest cause of primary amenorrhoea.
17. MRKH: ovaries always functional.
18. MRKH renal anomalies ≈ one-third, type B.
19. Dilation is first-line neovagina therapy.
20. Ileal neovagina stenoses more than sigmoid.
21. Bowel neovagina: adenocarcinoma; skin: squamous carcinoma.
22. Cervical agenesis: avoid uterovaginal anastomosis.
23. HWWS: ipsilateral renal agenesis; no amenorrhoea.
24. Vaginal rhabdomyosarcoma: grapelike mass, under 2 years.
25. Labial adhesions: 80% resolve within a year.
26. WHO type III FGM = infibulation.
27. Round ligament absent in genetic males.

59 Differences of Sexual Development

Reading time ≈ 36 min (slow pace)

1 · DEFINITION AND NOMENCLATURE

- **Differences (disorders) of sex development (DSD)**: chromosomal, gonadal or anatomical sex not typically male or female. Examples: congenital adrenal hyperplasia (CAH), ovotesticular DSD (OVT DSD), androgen insensitivity syndrome (AIS).
- **2006 Chicago consensus** (Lee et al.) replaced "intersex", "hermaphrodite" and "sex reversal" with karyotype-based terms; "atypical" preferred to "ambiguous" genitalia.

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not AIS) MCQ

Offer endocrine and genetic testing for **proximal hypospadias even with descended testes** — a named diagnosis predicts gender identity data, fertility, tumour risk, comorbidities and recurrence.

2 · NORMAL SEX DEVELOPMENT

Three steps: **chromosomal sex** (fixed at fertilisation) → **gonadal sex** (bipotential gonad until ≈ 6 weeks) → **phenotypic sex** (bipotential ducts and external genitalia until ≈ 8 weeks).

- **SRY** (sex-determining region Y): earliest testis-determining transcription factor; upregulates **SOX9** → Sertoli differentiation.
- Genes act elsewhere too, explaining syndromic DSD: **WT1** (kidney — Wilms, nephropathy), **SF1** (NR5A1) and **DAX1** (adrenal — adrenal failure), SOX9 (bone — campomelic dysplasia). DAX1 duplication (Xp21) causes 46,XY dysgenesis.
- Ovary needs RSPO1/WNT4/β-catenin and FOXL2; ovarian germ cells enter meiosis in fetal life, testicular ones only at puberty.
- **Sertoli → anti-Müllerian hormone (AMH)** at 7–8 weeks → Müllerian regression by ≈ 10 weeks.
- **Leydig → testosterone** from ≈ 9 weeks (autonomous, then placental human chorionic gonadotrophin [hCG], then fetal luteinising hormone [LH]) → Wolffian ducts → epididymis, vas, seminal vesicles.
- **5α-reductase type 2 → dihydrotestosterone (DHT)**, higher receptor affinity; DHT is the active androgen in prostate, urogenital sinus and external genitalia. External virilisation complete by 12–13 weeks; penile growth and descent in third trimester (androgen, INSL3).
- Female: no AMH → tubes, uterus, upper vagina; distal one-third of vagina from urogenital sinus; no androgen → clitoris, labia.

> **KEY POINT** The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. Any DSD phenotype can be reasoned from which hormone was missing, where and when — e.g. a uterus in 46,XY means Sertoli (AMH) failure, i.e. dysgenesis, not a synthesis or receptor defect.

3 · CLASSIFICATION (2006 / CHICAGO)

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY mixed gonadal dysgenesis; 46,XX/46,XY chimerism

Category	Conditions
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: CAH (21-hydroxylase, 11β-hydroxylase, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, androgenic drugs
46,XX — other	Cloacal anomalies; Müllerian agenesis (Mayer–Rokitansky–Küster–Hauser, MRKH); vaginal atresia
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular DSD; testis regression
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	Complete/partial AIS (CAIS, PAIS)
46,XY — other	Persistent Müllerian duct syndrome (PMDS); vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

3β-hydroxysteroid dehydrogenase (3β-HSD) and P450 oxidoreductase virilise 46,XX yet undervirilise 46,XY. 21-hydroxylase deficiency causes no genital difference in 46,XY.

4 · 46,XX DSD

Congenital adrenal hyperplasia — 21-hydroxylase deficiency

- CAH = commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-hydroxylase deficiency (21-OHD) = **95%** of CAH; autosomal recessive, ≈ 1 in 15,000.
- Mechanism: cortisol deficiency → ↑adrenocorticotropic hormone (ACTH) → adrenal androgen excess → virilisation; aldosterone deficiency → **salt wasting**.
- Infant presentation: variable virilisation, risk of adrenal crisis in the first weeks; infant death in the family suggests CAH.
- 46,XX: normal ovaries and Müllerian structures; clitoromegaly to fused labioscrotal folds, often a **common urogenital sinus**; may look like a boy with **bilateral impalpable gonads**. Graded by **Prader I–V** (increasing masculinisation; † V = male-appearing phallus with meatus at tip and fused empty scrotum).
- Diagnosis: ↑**17-hydroxyprogesterone (17-OHP)**, measured after **36 h** (physiological neonatal surge). Also ↑androstenedione, testosterone, renin; ↓cortisol; cosyntropin test.
- **Salt-wasting crisis**: hyponatraemia, hyperkalaemia, acidosis, hypotension, shock — usually not before **day 5–7**, so normal early electrolytes do not exclude CAH. Treatment: glucocorticoid + mineralocorticoid + salt — lifesaving († hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone and sodium chloride supplements in infancy; double-to-triple hydrocortisone during illness or surgery).
- 46,XY boys have normal genitalia, so **newborn screening** matters most for them.
- Long-term: lifelong replacement balancing androgen suppression against steroid excess (growth, obesity, bone). Cortisol's circadian and stress responses are hard to mimic, and androgens also arise by the "backdoor" pathway, so control is difficult. Fertility preserved but reduced. 46,XY: **testicular adrenal rest tumours** — benign, commoner with poor suppression, regress with more glucocorticoid; they obstruct tubules (azoospermia), so bank sperm; excision rarely indicated. Poor suppression can also cause hypogonadotropic hypogonadism.
- Prenatal dexamethasone must start by 6–7 weeks; ≥ 4 fetuses treated per girl helped — experimental (ethics-board-approved protocol only).

Form	Features
Classic salt-wasting	Cortisol + aldosterone deficient; crisis in first weeks
Simple virilising	Androgen excess, no infant shock

Form	Features
Nonclassic	20–50% enzyme activity; premature adrenarche, hirsutism, oligomenorrhoea; may need no treatment

Enzyme	Share	blood pressure (BP) / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; ↓BP, ↑K^+	Normal	↑17-OHP
11β-hydroxylase	≈ 5%	Hypertension, ↓K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	↑dehydroepiandrosterone (DHEA)
17α-hydroxylase	Rare	Hypertension, ↓K^+	Female/atypical; delayed puberty	↓androgens

Other 46,XX DSD

- **46,XX testicular DSD**: SRY translocated to X → testes, male genitalia, no Müllerian structures; azoospermia (no AZF region).
- Aromatase (CYP19) deficiency virilises mother and infant; maternal virilisation resolves after delivery but recurs in later pregnancies. Maternal androgens / aromatase deficiency: exposure ends at birth → no postnatal medical therapy.

5 · 46,XY DSD

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Bilateral streaks, female genitalia, uterus present; absent puberty; tumour 30–45% → prophylactic gonadectomy; lifelong oestrogen; pregnancy with donor ovum
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian structures; identity variable; WT1 → Denys–Drash (congenital nephrotic syndrome, Wilms 90%) / Frasier (late focal segmental glomerulosclerosis, gonadoblastoma 60%)
CAIS	X-linked receptor defect; female genitalia, short blind vagina, no uterus, inguinal/abdominal testes; breasts by aromatisation; ≈ 50% have inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia, residual receptor function; identity mostly male; pubertal breast growth
5α-reductase 2 deficiency	Autosomal recessive; atypical/female at birth, blind vaginal pouch; marked **virilisation at puberty**, fertility reported; testosterone:DHT **>20:1**; Dominican "guevedoces" (Imperato-McGinley 1979)
17β-HSD3 deficiency	↑androstenedione:testosterone; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH or AMH-receptor mutation (autosomal recessive); normal male genitalia + uterus/tubes; found at hernia repair/orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester → male genitalia, impalpable testes, no Müllerian structures; earlier loss → atypical genitalia. Undetectable testosterone and AMH, ↑gonadotropins at mini-puberty (1–6 months); testosterone from adolescence; testicular prostheses later

- **Synthesis/action defects**: AMH normal → no uterus; adrenal failure only if enzyme shared with cortisol pathway.
- **PAIS management**: raised male — high-dose testosterone may improve genital development; female identity — gonadotropin-releasing hormone agonist or gonadectomy, then oestrogen. CAIS:

vaginal dilation usually suffices; vaginoplasty rarely needed.

- **CAIS hernia**: in a girl at herniorrhaphy, vaginoscopy for cervix or look for a testis in the sac.
- **PMDS (Clarnette)**: 1 bilateral intra-abdominal (60–70%); 2 one testis in hernia sac/scrotum + contralateral hernia (20–30%, "hernia uteri inguinale"); 3 transverse testicular ectopia (10%). Orchidopexy; vasa hug the uterus, so remnants traditionally kept; sequelae: infertility, seminoma (intra-abdominal cryptorchidism), Müllerian malignancy 3–8% → discuss removal vs ultrasound or magnetic resonance imaging (MRI) surveillance; if removing, laparoscopic/robotic to spare vasa.

6 · SEX CHROMOSOME DSD AND OVOTESTICULAR DSD

Feature	Klinefelter (47,XXY)	Turner (45,X and variants)
Frequency	1 in 600 males; commonest aneuploidy	25–50 per 100,000 females
Phenotype	Tall; small firm testes <3.5 cm; gynaecomastia; neurodevelopmental issues	Short; female genitalia; absent puberty; webbed neck; coarctation, bicuspid valve; horseshoe kidney (renal anomaly ⅓–½)
Hormones	↑gonadotropins, ↓testosterone	Ovarian insufficiency ≈ 95%
Treatment	Testosterone (not in infancy); check gonadotropins and testosterone at mini-puberty; tubules hyalinise in adolescence; sperm extraction + intracytoplasmic sperm injection	Cardiac and renal screening at diagnosis, then lifelong; growth hormone; oestrogen + progesterone; oocyte donation (higher pregnancy risk) or oocyte cryopreservation
Tumour	Extragonadal germ cell; male breast	None without Y; **Turner + Y** 12–20% (gonadoblastoma → dysgerminoma); 2017 guideline: gonadectomy at diagnosis (low-level evidence, now challenged)

- **Mixed gonadal dysgenesis (MGD)**, usually 45,X/46,XY: dysgenetic testis + contralateral streak + Müllerian structures. Classic **asymmetry**: one palpable testis, impalpable side with flat labioscrotal fold, hemiuterus/tube on streak side (local AMH action). **Second commonest cause** of newborn atypical genitalia. Germ cell neoplasia 15–35% → remove streaks. All 45,X/46,XY need Turner screening (cardiac, renal, growth). ≈ 85% of 45,X/46,XY have male-typical genitalia. Remaining testis: raised male — keep descended testis, bring down intra-abdominal one; raised female — remove or relocate to groin/labioscrotal fold for surveillance. Gender dysphoria in a substantial minority raised female.
- **Ovotesticular DSD**: ovarian + testicular tissue (often ovotestis); mostly 46,XX; urogenital sinus and uterus usual. A vas always accompanies a testis, a tube an ovary. Ovary usually normal, testis dysgenetic; tumour ≈ 3%; polar biopsy for bipolar ovotestis (patterns: admixed, compartmentalised, bipolar). Discordant tissue is hard to remove, so irreversible steps are deferred; puberty follows the functioning tissue, usually oestrogenic.

7 · EVALUATION OF THE NEWBORN WITH ATYPICAL GENITALIA

> **KEY POINT** **Emergency**: atypical genitalia with bilateral impalpable gonads = salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP; watch for crisis from the end of week 1. Everything else (designation, surgery) is not an emergency.

- Multidisciplinary review within **24–48 h**; congratulate, say "your baby", never "it"; gender-neutral terms. A urologist without a team should transfer early.
- History: infertility, amenorrhoea, unexplained infant death (CAH), maternal steroids/progestins.
- Examine: stretched phallic length, glans width, orifices, labioscrotal fusion and rugation, anogenital distance, **gonads**.
- **Gonads are the most informative sign**: ovaries do not descend, so a palpable gonad = testicular tissue (excludes typical CAH). 0 palpable → CAH; 1 → MGD, OVT DSD, PAIS, partial dysgenesis; 2 → 46,XY DSD.

- Hypospadias + undescended testis without atypical genitalia (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if proximal meatus.

ALGORITHM
Karyotype (≥45 cells; fluorescence in situ hybridisation for X/Y in hours) + electrolytes + 17-OHP at ≈ 36 h → pelvic ultrasound (US) for uterus (best neonatally) and gonads → at ≈ 1 week: LH, follicle-stimulating hormone (FSH), testosterone, oestrogen, androstenedione, DHT, AMH ± hCG stimulation → genetic panel / microarray / exome → if unclear: MRI, genitography, endoscopy, laparoscopy ± gonadal biopsy (**no gonadectomy during diagnosis**)

Test	Interpretation
17-OHP	Raised in CAH (21-OHD)
AMH	Normal → functioning Sertoli (or granulosa) cells
LH, FSH	Raised → gonadal failure
Androstenedione / DHEA sulfate	21-OHD, 17β-HSD3 / 21-OHD, 3β-HSD
Testosterone:DHT	>20:1 → 5α-reductase deficiency
hCG stimulation	No testosterone rise → no testis, Leydig hypoplasia or LH receptor defect

Karyotype	Findings	Likely
46,XX	Impalpable gonads, uterus, ↑17-OHP	CAH
46,XX	Gonads not two ovaries	OVT DSD
46,XY	Uterus ±	Partial dysgenesis, OVT DSD
46,XY	No uterus	Androgen synthesis/action defect
45,X/46,XY	Asymmetric gonads, uterus ±	MGD

Precise diagnosis is nearly always reached in 46,XX DSD but in **<50%** of 46,XY DSD. Late presenters (typical external genitalia): CAIS, Swyer, MRKH, Turner + Y, Klinefelter, 5α-reductase deficiency; basal hormones are informative after puberty (↑gonadotropins + ↓sex steroid → gonadal failure; ↑LH + ↑testosterone with female genitalia → CAIS). Examine adolescents patiently, with few clinicians, only when management may change.

8 · GENDER ASSIGNMENT AND THE MULTIDISCIPLINARY TEAM

- Team: endocrinologist, urologist/surgeon, gynaecologist, geneticist, psychologist/psychiatrist, social worker, ethicist, nurse, peer support. **Same-day joint clinic**; lifelong care with planned transition.
- Historical **Money (1975)** model — early designation, early surgery, secrecy — failed; replaced by transparency and shared decision-making.
- Designation weighs diagnosis-specific identity data, prenatal androgen exposure, sexual function, fertility, hormonal potential and family/culture. Sex of rearing predicts satisfaction better than androgen exposure; no test predicts identity.
- Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male trend: PAIS, 5α-reductase and 17β-HSD3 deficiency, partial dysgenesis.
- Gender dysphoria highest in 5α-reductase deficiency raised female (≈ half); lowest in 46,XX CAH raised female.
- Micropenis (stretched length <2.5 SD; <2 cm at 0–5 months) → endocrine work-up; female reassignment no longer recommended.

9 · SURGERY

- **Feminising genitoplasty**: clitoroplasty, vaginoplasty/urogenital sinus mobilisation, labioplasty → *Ch 60*. Clitoral surgery is irreversible and can be catastrophic if identity later differs; benefits of

infant surgery unproven. Only urgent indication: urinary retention/obstruction.

- **Masculinising**: mainly proximal hypospadias repair ± penoscrotal transposition, orchidopexy; high complication rates; largely revisable, so infant repair often chosen when raised male.
- **Timing debate**: ideally defer irreversible surgery until the individual can participate; parents often prefer early surgery. CAH surgery is debated separately.
- MRKH (46,XX, absent uterus and upper vagina, normal ovaries; upper tract anomaly ≈ ⅓ → renal US): dilation first-line; symptomatic uterine remnants are removed or suppressed — never joined to a neovagina (no cervix); Vecchietti; buccal or bowel vaginoplasty; uterus transplantation investigational.

10 · GONADAL TUMOUR RISK AND GONADECTOMY

Risk rises with Y material (gonadoblastoma locus, TSPY), dysgenesis and intra-abdominal position; falls with a scrotal gonad. Shift from automatic gonadectomy to individualised risk, function and autonomy (biopsy, relocation, tissue cryopreservation).

Risk	Condition	Germ cell tumour	Management
High	Gonadal dysgenesis + Y, intra-abdominal (incl. Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal gonad	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (now debated)
	17β-HSD3 deficiency	28%	Monitor or gonadectomy
	PAIS / dysgenesis, scrotal gonad	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4 staining)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove testicular tissue if discordant
?None	5α-reductase deficiency; Leydig hypoplasia	Unknown	Unresolved

11 · ETHICS

- Principles: bodily autonomy, transparency about known and unknown, avoiding irreversible procedures before consent unless medically necessary, informed parental consent covering limits of predicting identity and fertility.
- Psychosocial morbidity (anxiety, depression, suicidality) is higher → integral psychology input; patient preference for terms (often the diagnosis itself) respected.

12 · MCQ ONE-LINERS

1. "True hermaphrodite" is now ovotesticular DSD.
2. "XY sex reversal" = 46,XY complete gonadal dysgenesis.
3. SRY upregulates SOX9 to drive Sertoli differentiation.
4. Sertoli cells secrete AMH from 7–8 weeks.
5. DHT is the active androgen for external genitalia and prostate.
6. Commonest cause of neonatal atypical genitalia: CAH (21-hydroxylase).
7. Second commonest: mixed gonadal dysgenesis (45,X/46,XY).
8. Measure 17-OHP after 36 hours of age.
9. Salt-wasting crisis usually occurs after day 5–7.
10. 11β-hydroxylase deficiency causes hypertension and hypokalaemia.

1. 17α-hydroxylase deficiency: hypokalaemia, not hyperkalaemia.
2. 3β-HSD deficiency undervirilises 46,XY and virilises 46,XX.
3. Uterus in 46,XY points to gonadal dysgenesis.
4. Testosterone:DHT >20:1 suggests 5α-reductase deficiency.
5. Guevedoces: 5α-reductase deficiency virilising at puberty.
6. CAIS gonads can stay until after breast development (risk 2%).
7. 1–2% of girls with inguinal hernia have CAIS.
8. Frasier syndrome: gonadoblastoma 60%, no Wilms predisposition.
9. Denys–Drash: Wilms 90%, congenital nephrotic syndrome.
10. PMDS: AMH or AMH receptor mutation; hernia uteri inguinale.
11. PMDS underlies 30–50% of transverse testicular ectopia.
22. 46,XX testicular DSD: SRY translocation; azoospermic.
23. Palpable gonad = testicular tissue; ovaries do not descend.
24. Klinefelter: commonest sex chromosome aneuploidy, 1 in 600 males.
25. All 45,X/46,XY patients need Turner screening.
26. Vas always accompanies a testis in ovotesticular DSD.
27. Precise diagnosis reached in <50% of 46,XY DSD.
28. No gonadectomy during the diagnostic phase.
29. Micropenis: stretched length <2.5 SD below mean.
30. Highest dysphoria: 5α-reductase deficiency raised female.
31. MRKH uterine remnant must never be connected to a neovagina.
32. Testicular adrenal rest tumours regress with increased glucocorticoid.
33. Pubertal gonadal biopsy: ≥30 tubules with OCT3/4 staining.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

Reading time ≈ 44 min (slow pace)

1 · PRINCIPLES

- Most children with **differences (disorders) of sex development (DSD)** have genital atypia. Diagnosis, karyotype, **congenital adrenal hyperplasia (CAH)** medical care and gender assignment → *Ch 59*.
- The controversy is **not technique but timing, and who decides** (parents or child). Present pros and cons of surgery *and* no surgery; never talk anyone into surgery; support the family whichever path is chosen; give access to all support groups.
- **Team**: family, paediatric urologist/surgeon, endocrinologist, geneticist, neonatologist, psychologist/psychiatrist, ethicist. The child is not named until evaluation is complete.

2 · UROGENITAL SINUS AND CLOACA — CLASSIFICATION

A **urogenital sinus** is a persistent communication of the vagina with the urinary tract (anywhere from meatus to bladder, usually mid–distal urethra); both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly CAH
Pure urogenital sinus	Normal external genitalia; behaves anatomically like a cloaca
Persistent cloaca	Rectum also joins: **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH: commonest cause of newborn atypical genitalia — half of all and **90%** of 46,XX cases; salt wasting in **67–75%**; karyotype within **48 h**.
- Cloaca: ≈ 1 in 20,000–50,000; the most challenging **anorectal malformation (ARM)**, 5–13.6% of ARMs.

> **KEY POINT** **Governing principle**: **true urethral length** (bladder neck → confluence) decides the operation far more than common-channel length — in sinus and cloaca alike. Confluence level is a continuum, not simply "high" (suprasphincteric) vs "low".

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants); virilisation lengthens the **vagina-to-meatus** distance, so "high" CAH confluence is an illusion of virilisation, reinforced by a male-like sphincter and verumontanum-like confluence.
- **Rink classification**: common-channel length, bladder-neck-to-vagina distance, confluence level, clitoral size, appearance.

3 · CLINICAL EVALUATION

- **History**: antenatal ultrasonography (US) shows fluid-filled pelvic structures; maternal androgens; family **early infant death** or electrolyte disturbance (CAH); siblings with atypia.
- **First threat is metabolic**: salt wasting (hyponatraemia, hyperkalaemia, high urinary sodium, raised renin) → adrenal crisis. Monitor electrolytes before thinking of reconstruction.
- **Blood pressure**: hypertension → 11β-hydroxylase deficiency.
- **Suprapubic mass**: bladder and/or **hydrometrocolpos** — often the only early sign of pure urogenital sinus.
- **Spine**: flat buttocks/abnormal crease (sacral agenesis) are commoner clues than skin markers.
- **Clitoris**: **Huffman index** (glans width × length) normal **<3.5 mm²**, concern **>10 mm²**. Isolated clitoral hypertrophy in a preterm infant is not a DSD.

- **Gonads**: **both descended → 46,XX extremely unlikely**. Labioscrotal hyperpigmentation in CAH (melanocyte-stimulating hormone).
- **Anus**: anterior displacement bridges to cloaca. **Posterior cloaca**: sinus opens into the anus (type A) or just anterior (type B).
- **Cloaca**: single orifice; from near-normal female to a blank "doll-like" perineum or phallus-like structure.

Hydrocolpos in cloaca: urine flows preferentially into the poorly draining vagina(s) on voiding, plus maternal-oestrogen-driven cervical mucus. Commoner with common channel **>3 cm**, duplication and posterior cloaca. Severe distension → respiratory distress, cyanotic legs, acidosis.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, ectopia	Majority
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on magnetic resonance imaging (MRI) ≤ 43%
VACTERL (vertebral, anal, cardiac, tracheo-oesophageal, renal, limb); gastrointestinal	Tracheo-oesophageal fistula; malrotation, duodenal atresia	Correct malrotation at colostomy

> **KEY POINT** Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). The number of perineal orifices tells you which world you are in.

4 · IMAGING AND ENDOSCOPY

Modality	CAH sinus	Pure sinus / cloaca
Ultrasonography	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as computed tomography (CT) or MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- **Genitogram**: Foley balloon occludes meatus → contrast → images on voiding. **Cervical impression** at vaginal dome = female internal organs.
- **Gonadal biopsy** (laparoscopic) only if it would change gender of rearing; **deep longitudinal incision**, because ovarian tissue may surround the testicular part of an ovotestis.
- **Cloaca prenatally** (from 19 weeks): transient fetal ascites (urine refluxing through uterus and tubes) + bilobed pelvic cysts + bilateral hydronephrosis + reduced liquor.
- Plain film: linear/meconium calcifications from retrograde urine flow. Hydronephrosis usually from hydrocolpos compressing the bladder neck. **Echocardiography always.**
- **Cloacogram**: balloon catheters in bladder, vagina(s), rectum → CT/MRI contrast → three-dimensional reconstruction. Targets: confluences, urethral length, vaginal anatomy, true rectum (may lie well above the fistula).
- Duplicated vaginas (≈ half) usually lie side by side with a cervix each; rectal fistula commonest **within the vaginal septum**.
- **Early endoscopy**: the distended vagina compresses the bladder anteriorly — **empty the vagina first** to reach the bladder.

Common channel (Peña)	<3 cm	>3 cm
Anomalies; hydrocolpos	Fewer	More (urological defects nearly all)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often

Urethral-type (narrow) channels raise outlet resistance → hydrocolpos; vaginal-type (redundant) channels do not.

5 · DSD RECONSTRUCTION — TIMING AND CONSENT

- **For early surgery** (historically 2–6 months): pliable tissue, less psychological trauma, parental relief.
- **Against**: advocacy and human-rights bodies seek a moratorium until the patient can consent; eventual gender identity is unknown.
- Surgery does not "cure" a DSD; the brain is the dominant sexual organ. Infant outcomes take ≈ 20 years, so published results may describe abandoned techniques.
- Most CAH patients and parents support parental decision; regret is never severe. All families meet a psychologist; observation with support is always offered.
- **Clitoroplasty**: only for severe hypertrophy (**Prader III–IV**) — the most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral and sinus tissue (lost once mobilised); maternal oestrogen thickens tissue; planned one-stage repair does better
After puberty	Infant stenosis; avoids dilation; consent; delay the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

> **KEY POINT** Oestrogen is invoked by both camps (maternal vs pubertal). With no comparative data, the decision is ethical rather than evidential; it currently rests with parents and the team. The Endocrine Society CAH guideline (2018) leaves it to families with experienced surgeons; the 2006 consensus limits surgery to Prader III–IV.

Body	Position
Lawson Wilkins Pediatric Endocrine Society / European Society for Paediatric Endocrinology CAH group (2002)	Surgery at 2–6 months; not between 12 months and adolescence
International Society of Hypospadias and DSD poll	Most surgeons prefer single-stage surgery before 2 years
American Medical Association ethics council	Rejected a moratorium after 2 years' review
Advocacy groups; some US state bills	Moratorium until consent; parental decision seen as a rights violation; bills not enacted

6 · OPERATIVE PRINCIPLES

Correct clitoral hypertrophy, fused anterior labia majora, absent labia minora and the common sinus: **clitoroplasty → vaginoplasty → labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession with denuded glans	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving neurovascular bundle	Basis of later techniques

Clitoral surgery	Technique	Verdict
Kogan (1983)	Subtunical excision via lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares dorsal bundle
Pippi Salle (2007)	Corpora disassembled, buried in labia	Reversible if male identity emerges

- **Fortunoff (1964)** posteriorly based perineal flap underlies almost every repair; now **omega-shaped**.
- **Hendren and Crawford (1969)**: pull-through; **confluence level determines vaginoplasty type**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, perineal flap inset; confluence unmoved	**Never for high confluence**: vaginal voiding, infection, incontinence
Pull-through	Very high	Vagina separated; sinus becomes urethra	Vagina falls short → flaps
Replacement	Absent vagina	Several → *Ch 58*	—

KEY POINT Ask: **must the confluence be moved?** A flap brings skin to a low vagina; a pull-through takes a high vagina off the urethra. A flap on a high confluence is the classic error.

7 · LOW CONFLUENCE REPAIR

ALGORITHM
Metabolic stability → **stress-dose steroids in CAH** → antibiotics → endoscopy → **Fogarty balloon in vagina** + Foley in bladder → total lower-body preparation (allows supine–prone rotation; lithotomy blocks conversion)

ALGORITHM
Glans traction suture → incision around glans, inner prepuce kept for hood (**second only to glans in sensitivity**) → strips either side of the ventral mucosal plate → omega flap → Y-incisions on labia majora → deglove to bifurcation, ventral plate intact

- **Nerves** run along the ischiopubic rami → paired dorsal bundles → glans; arteries (internal pudendal) run medially on the bifurcated corpora, then dorsally.
- **Ventral subtunical excision (authors)**: tourniquet → incisions through Buck fascia on the **ventral-most** corpora → erectile tissue teased out → tunics and dorsal bundle untouched → stumps ligated. Mobilising the bundle risks sensation and vascularity.
- Alternatives: **bundle mobilisation** (if done, dissect beneath the second layer of Buck fascia, just on the tunica albuginea); **reapproximation** (middle erectile tissue excised, corpora rejoined); **corporal preservation** (corpora split and fixed to the pubis — no long-term data); **disassembly** (reversible).
- **Glans reduction**, if any, ventral midline (perforating nerves enter dorsally); never excise glanular epithelium. Secure glans to corporal stumps, not the pubis (sits too high).

ALGORITHM
Flap vaginoplasty: omega flap with fat → posterior sinus dissected off rectum (**hardest step**) → posterior sinus opened in midline into posterior vaginal wall → **continue to normal-calibre vagina** → flap inset

The flap must be tension-free, wide enough for a normal introitus, and not redundant (obstructing mound). **Labioplasty**: clitoral skin split as **labia minora** (Marberger); labia majora moved inferiorly by **YV-plasty** so the introitus lies between them.

> **KEY POINT** Two fixed points: never touch the dorsal neurovascular bundle, and always open back to normal-calibre vagina.

8 · HIGH CONFLUENCE: PULL-THROUGH

- Only ≈ **5%** of CAH; commoner in pure sinus.
- **Critical step**: separating the anterior vaginal wall from urethra and bladder neck — no plane, sphincter and nerves at risk, hardest to see. Poor exposure → stricture, fistula, diverticulum.
- **Passerini-Glazel flap**: sinus tubularised with clitoral skin to cover the separation and build the anterior vaginal wall. **Gonzalez–Fernandes**: preputial flap for vestibule and anterior wall.
- **Anterior sagittal transanorectal approach (ASTRA)**: prone, anterior rectal wall divided — excellent exposure, no colostomy, continence unaffected.

ALGORITHM
Prone → midline dissection between sinus and rectum → sinus divided to normal vagina → retractor inside vagina → anterior wall dissected off urethra **under direct vision** → sinus closed as urethra in 2–3 layers → vagina mobilised → supine: flaps + labioplasty

At the separation **err on the side of the vagina**. The vagina often falls short, so skin flaps are brought up; sinus tissue is more reliable. For a very small, very high vagina, consider waiting until puberty.

9 · TOTAL AND PARTIAL UROGENITAL MOBILISATION

- **Total urogenital mobilisation (TUM, Peña 1997)**: devised for cloaca; whole sinus dissected circumferentially above the pubis (dividing **avascular pubic ligaments**) and brought to the perineum, avoiding separation. Less fistula and stenosis, better cosmesis.
- TUM **enables** a vaginoplasty rather than replacing it; the highest vaginas still need pull-through, made easier.
- **Never discard the sinus**: split **ventrally** = mucosa-lined vestibule (flap repair); split **dorsally** = anterior vaginal wall (pull-through).

	TUM	**Partial (PUM, Rink)**
Extent	Retropubic and suprapubic	Stops at **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · OUTCOMES OF SINUS/DSD SURGERY

- Evidence: small retrospective series, surrogate endpoints (patency, appearance) instead of sensation, orgasm and painless intercourse.
- Infant vaginoplasty usually needs a **secondary introitoplasty** after puberty (**25–100%**) — the orifice does not grow because nothing passes through it; revision is easy.
- **Azziz (1986)** MCQ: poor coitus rates, but almost all had **cut-back** — not applicable to modern repairs.
- Early single-stage repair → less stenosis than late vaginoplasty; planned one-stage beats planned multistage. **Confluence level, number of procedures and surgeon expertise** outweigh age at surgery. No data show better results after postpubertal surgery.
- Stenosis ranges from a minority to over a third, highest where cut-backs predominated; dyspareunia is common even without stenosis.

- Clitoroplasty preserves nerve conduction, but conduction is not sensation; orgasm does not differ between operated and non-operated women. Physicians rate appearance better than patients.
- Genitosensory testing shows altered hot, cold and vibratory sensation in some series; cosmesis is unsatisfactory in up to half in older series; no devascularisation after modern clitoroplasty.
- Quality of life and mental health are normal in most; relationships begin later. Lower urinary tract symptoms do not differ from controls.
- Sexual satisfaction is as in controls except **salt-wasting CAH**; feminising surgery neither improves nor impairs psychosexual outcome.
- **Elmasri meta-analysis (2018)**: observational only; mean age at surgery 2.7 years; female gender identity ≈ 89%; vaginal stenosis **27%**; only half comfortable intercourse; no early-vs-late comparison.

Two-procedure dilemma	Infancy	After puberty
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations; a moratorium would ensure the question is never answered. The most important factor in results is a **surgeon/team with special interest** — the case for centres of excellence.

11 · CLOACA — INITIAL MANAGEMENT

- **Hendren's rule**: rectal pull-through should never be a separate procedure; repair rectum, vagina and urethra in one stage (exposure, untouched tissue, no rectal remobilisation). Some groups stage (anorectal first, urogenital peripubertally).
- **Neonatal pitfalls**: unmanaged hydrocolpos, stoma problems, misdiagnosis.

ALGORITHM
(1) Decompress bowel → (2) decompress genitourinary tract → (3) correct nephron-destructive anomalies → (4) definitive repair

Stabilise first: abdominal distension may compromise respiration. Definitive repair is never the first operation.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon for vaginal replacement	Proximal site absorbs urine → acidosis
Descending, divided (Levitt, Peña)	Less mucosa for resorption; enough distal colon; less prolapse	Leave a long sigmoid

- Colostomy siting trades distal colon length (pull-through, vaginal replacement) against urine resorption through colonic mucosa (**hyperchloraemic acidosis**) and prolapse. Correct malrotation; **preserve the appendix** for later appendicocaecostomy.
- **Urinary decompression**: voiding into vagina/rectum → hydronephrosis, urinary tract infection (UTI), acidosis. **CIC first line** (even into the vagina) → cloacal cut-back → vaginostomy/vesicostomy only if CIC fails (vaginostomy may tether the vagina).
- At colostomy, endoscopy drains and irrigates the vagina (may reveal the bladder), clears inspissated mucus from the fistula, and leaves feeding tubes in all channels for contrast studies.
- **Kidneys**: structural anomalies **60–83%**; chronic kidney disease (CKD) **75%**, end-stage renal disease (ESRD) **17%**; treat obstruction promptly. Urodynamics abnormal in **90%** (overactivity, atonia, dyssynergia, high leak point pressure). Reflux is managed medically if UTI is prevented.

12 · CLOACA — DEFINITIVE REPAIR

At **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Transabdominal TUM never (ischaemia)

KEY POINT Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags it below the continence mechanism → leakage.

ALGORITHM
Posterior sagittal anorectovaginourethroplasty (PSARVUP, Peña): coloured catheters in each channel → **prone jackknife** → muscle stimulator marks anal site → midline incision → rectal fistula opened, rectum separated and mobilised to reach skin → TUM or vaginal separation → urethra closed in 2–3 layers + **fat flap** → perineal body → rectum centred in sphincter mass

- Rectum and vagina share a wall: **better to enter the vagina than the rectum or urethra**. Rectum above the pubococcygeal line needs laparoscopy or laparotomy.
- Vaginal separation (channel >3 cm or urethra <1.5 cm) is more tedious than rectal separation; entering the urinary tract is an error. Incise a side-by-side septum at initial endoscopy. The vaginal "switch" and separating a vagina that falls short after TUM are abandoned (ischaemia, stenosis).
- Technique: **sharp dissection, not cautery**; the vagina often **almost encircles the urethra**. Very high or absent vagina → bowel interposition → *Ch 58*.
- Aftercare: Foley 2 weeks; rectal dilation from 2–3 weeks; colostomy closure ≈ 3 months. Poor emptying in almost one-third → CIC at any doubt.

13 · CLOACA — RESULTS AND FOLLOW-UP

- Outcome depends mostly on the **spine** (short cord worst) and **common-channel length**.
- **Urinary**: about half void spontaneously; Peña: CIC **28%** (channel <3 cm) vs **78%** (>3 cm). Lifelong yearly urological review.
- Good voiding signs: good bladder neck, short channel, normal sacrum, two normal kidneys. Neuropathy far commoner with channel >3.5 cm; many have leak point pressure >40 cm H_2O despite a normal spinal MRI.
- **Renal**: CKD mostly stage I–II; preserved by aggressive bladder management (CIC, augmentation, diversion).
- **Faecal**: tracks neurology (Hendren's good signs: perineal raphe, anal dimple, normal spine, brisk muscle reflex). Most have voluntary movements but some soil; constipation common; continence improves into adulthood. Bowel programme, then **Malone antegrade continence enema (MACE)**.
- **Gynaecological**: introital stenosis, retained septum, persistent fistula, deficient perineal body (often from an anteriorly placed rectum); about half of outside repairs need redo; vaginal stenosis/obstructed menstruation **36–41%**; obstructive haematometra; pregnancy possible, **caesarean preferred**. Leave atretic genitalia — they may develop.

ALGORITHM
US **6–9 months after thelarche** → repeat every 6–9 months until menses are normal

14 · MCQ ONE-LINERS

1. True urethral length, not channel length, decides the operation.
2. CAH confluence: 1.5–2 cm from bladder neck.
3. Huffman index normal <3.5 mm^2.
4. Both gonads descended: 46,XX unlikely.

5. Hydrometrocolpos: often the only early sign of pure sinus.
6. Cerebriform adrenals suggest CAH.
7. Rectal fistula commonest in the vaginal septum.
8. Schmid (1961): first nerve-sparing clitoroplasty.
9. Flap vaginoplasty for high confluence: classic error.
0. Pull-through: Hendren and Crawford (1969).
1. PUM stops at the pubourethral ligament.
2. Hendren's rule: never a separate rectal pull-through.
3. CIC is first-line decompression in cloaca.
4. Cloaca: CKD 75%, ESRD 17%.
5. TUM in cloaca: channel <3 cm and urethra >1.5 cm.
6. Short spinal cord most impairs continence.
7. Four sinus entities: genital atypia, pure sinus, cloaca, female exstrophy.
8. Hypertension in a virilised infant: 11β-hydroxylase deficiency.
9. Cervical impression on genitogram indicates female internal organs.
20. Ovotestis biopsy: deep longitudinal incision.
21. Fetal ascites in cloaca: urine refluxing through uterus and tubes.
22. Tethered cord in ≈ one-third of cloacas, mostly without skin stigmata.
23. Clitoral erectile tissue removed through ventral incisions; dorsal bundle untouched.
24. Labia minora from split clitoral skin; labia majora moved by YV-plasty.
25. ASTRA divides the anterior rectal wall; no colostomy needed.
26. Mobilised sinus split ventrally = vestibule; dorsally = anterior vaginal wall.
27. Azziz series: poor results reflect cut-back vaginoplasty.
28. Cloaca colostomy of choice: divided descending colostomy.
29. Preserve the appendix at colostomy for later MACE.
30. Transabdominal TUM never: channel ischaemia.
31. Ultrasound for menstrual obstruction 6–9 months after thelarche.

65 Adolescent and Transitional Urology

Reading time ≈ 14 min (slow pace)

1 · DEFINITIONS

- **Adolescent urology**: the bridge between paediatric and adult urology for patients with complex congenital conditions. The adult urologist has the operative skill but lacks **whole-patient knowledge**: multiple prior operations, altered anatomy and psychology, expert and demanding families, paediatric-formed expectations.
- **Transition** (Got Transition): planned move from child/family-centred to adult/patient-centred care, with or without a change of clinician — **planning → transfer → integration**. **Transfer** is only one step.
- Avoid the word "normal": the patient is their own baseline.

> **KEY POINT** Evidence for transition looks weak because the wrong endpoint is measured. Renal function and glycated haemoglobin (HbA1c) are expected to worsen, so **stability and retention in follow-up** are success; fewer patients are lost when actively transitioned.

2 · WHO NEEDS TRANSITION

Category	Examples	Implication
1. Complex congenital	Exstrophy, posterior urethral valves, complex hypospadias, spina bifida (SB)	Planned lifelong care
2. Treated, now quiescent	Vesicoureteric reflux	Early-adult review; may discharge with open return
3. Cured, with treatment sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Not every transitioned patient needs lifelong follow-up. **Failed transition**: poor understanding of the condition, overbearing family control, loss to follow-up, unrecognised complications, reliance on the emergency department (ED), lost social support → higher morbidity and mortality. Harm comes through **loss of preventive surveillance**.

3 · THE PROCESS

"Three Ps"	Content	Success measure
Preparation	Independence; usually from 12–14 years; Transition Readiness Assessment Questionnaire (TRAQ)	All families involved
Plan	Clinic, imaging, blood-test schedule; self-care (catheterisation, irrigation, medications, whom to call)	Plan shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- The **paediatric provider must initiate**; "completing" care and leaving the patient to find their own way is unacceptable.
- **Patient**: must be stable — avoid transfer during a crisis (declining renal function, urgent surgery). Exception: **pregnancy** → expedited transfer with paediatric–adult co-management.
- **Parents**: losing control and a trusted provider is a major barrier. Give **private time** for sexual function and debut, then ask what may be shared.
- Begin ≈ **12 years** (American Academy of Pediatrics, AAP). Most important skill: finding valid information. Complete by late teens–early 20s; **26** = latest (US parental insurance limit). If independence is unachievable, plan custodianship (e.g. sibling).

- Programmes: Good 2 Go (Toronto), Ready Steady Go (UK), Got Transition (USA).

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning (parallel)	14–18
Transfer of care	18–21
Transition complete (integration confirmed, feedback)	18–23

> **KEY POINT** Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · ROUTES AND MODELS OF CARE

- Routes: transitioned referral (ideal); referral without transition (suboptimal); **new adolescent problem (ideal** — managed from outset); **emergency after discharge** (most challenging, often most prevalent).

Model	Strength	Drawback
1a Lifelong paediatric urologist	Absolute continuity; suits limited life expectancy	Patient outlives provider; paediatric environment unsuitable
1b Paediatric urologist integrated with adult department	Continuity + adult expertise; joint clinics and operating	Cost, privileging, geography
2 Transfer to adolescent/adult specialist elsewhere	Specialist coordinates multidisciplinary team (MDT)	Cross-institution communication
3 Direct transfer, no plan	Default	Invisible until emergency

"Self-induced model 3": moving for college or work → lost follow-up → crisis. Lifelong need and multidisciplinary need are **independent**: exstrophy may need only a urologist lifelong; SB needs long-term MDT (neurosurgery, orthopaedics, colorectal, physiotherapy). **Nursing** is the service backbone.

5 · DEVELOPMENTAL CONSIDERATIONS AND CONSENT

- Surgical timing rests on weak mid-20th-century psychoanalytic theory. Phallic–oedipal period (**3–6 years**) → avoid genital surgery; first year → avoid separation.
- Genital surgery windows: both developmental theory and the 1975 AAP panel favour **<6 months**; second window differs (2–3 years vs 4th–5th year). Elective timing is driven by school, parental anxiety and surgeon availability; semi-elective (exstrophy closure) → as soon as safe.
- Augmentation cystoplasty withheld until the child can catheterise and wants dryness.
- Self-management (adherence, responsibility, skills, knowledge) matures into the third decade. **Emerging adults (18–25)** consent for themselves unless lacking capacity.
- **SB**: usually normal intelligence but deficits in **working memory and executive function** — critical to decisions; a trusted urologist and nurse practitioner help.
- **Medical post-traumatic stress disorder (PTSD)**: distrust, delayed care, avoidance, panic around procedures; cognitive-behavioural therapy (CBT) may help. Ask: could surgery have waited for the patient's consent?

6 · SETTING UP A SERVICE

- No defined training pathway; most providers train in paediatric and reconstructive urology.
- Core MDT: paediatric and adult urologists, specialist nurses, complex-care coordinator, nephrologist, psychologist; others as needed. Relationships determine long-term success.
- Institutional support via a **business model** plus patient feedback.
- Adult practice must explain: patient decides; consent and confidentiality reviewed **with family present and again alone**; electronic record communication; after-hours use; standing orders (e.g.

urine cultures) to avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear of unknown	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity (biggest)**; finance	Partnerships, networks, training

7 · MCQ ONE-LINERS

1. Transition: planning, transfer, integration.
2. Stability, not improvement, is success in progressive disease.
3. Vesicoureteric reflux: may be discharged after early adult review.
4. Pelvic rhabdomyosarcoma survivors: surveillance for treatment late effects.
5. Paediatric provider initiates transition.
6. Avoid transition during a medical crisis.
7. AAP: begin transition at about 12 years.
8. Age 26: latest commonly cited completion age.
9. Transfer of care: 18–21 years.
0. Transition complete by 23, after integration confirmed.
11. New adolescent problem is an ideal route.
12. Emergency post-discharge referral: most challenging route.
13. Model 1b combines continuity with adult expertise.
14. Most complex MDT needs: spina bifida.
15. Nursing is the backbone of transitional care.
16. Avoid genital surgery at 3–6 years.
17. Both schemes agree genital surgery before 6 months.
18. Spina bifida: deficits in working memory, executive function.
19. Confidentiality discussed with family, then alone.
20. Biggest adult-provider barrier: unfamiliarity with congenital conditions.

120 Aging and Geriatric Urology

Reading time ≈ 21 min (slow pace)

1 · CONCEPTS AND BIOLOGY

- Urology is a predominantly geriatric specialty. Age raises both the need for care and the risk of surgery and drugs, but **chronological age is a poor descriptor**: what matters is physiological and functional age, measured by function, frailty and geriatric syndromes. The United Nations proposes defining old age by years of life remaining.
- "Geriatrics" first used as a clinical term by Nascher (1909) MCQ.
- **Cellular senescence** (irreversible replication arrest) is the fundamental driver of ageing. Replicative senescence follows telomere shortening — the **Hayflick limit** (1961) MCQ; stress-induced senescence follows reactive oxygen species, radiation or DNA damage.
- **Senescence-associated secretory phenotype (SASP)**: cytokines, growth factors, matrix metalloproteinases → chronic inflammation, tumorigenesis; senescent cells age neighbouring tissue.
- **Healthy ageing** (World Health Organization, WHO): developing and maintaining the functional ability that enables wellbeing — intrinsic capacity interacting with environment. The target of care is function, not absence of disease.

2 · CLINICAL EVALUATION

- Four in five adults >65 have ≥1 chronic condition. Geriatric syndromes act **synergistically** (many interacting factors → one phenotype such as functional decline), unlike the linear one-cause model of trauma.
- **Homeostenosis**: age-related narrowing of physiological reserve, so balance is restored less well after a stressor (infection, surgery, chemotherapy).

Functional assessment

Tier	Content	Significance
Basic activities of daily living (BADLs/ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Critical to living independently

- Advanced activities of daily living form a third, higher tier. Function is lost top down; each step (one IADL → one BADL) shortens survival and raises cost. Screening question: **can the patient bathe independently?** Occupational therapy restores some independence; a baseline lets you monitor change with treatment.

Surgical risk — optimise, do not "clear"

- Age is seldom an independent predictor of surgical morbidity/mortality. Predictors: **longer operating time, poor American Society of Anesthesiologists (ASA) score, urgent surgery**. Optimisation reduces complications even >80. "Clearance" is a misnomer — the surgeon owns recognition of comorbidity and syndromes.

ALGORITHM
Frailty screen before elective major surgery (Society for Perioperative Assessment and Quality Improvement, SPAQI) → positive → diagnostic frailty assessment → comprehensive geriatric assessment (CGA)

- **American College of Surgeons National Surgical Quality Improvement Program (ACS NSQIP)/American Geriatrics Society (AGS) checklist**: 13 items — cognition/capacity, depression, delirium risk, substance use, cardiac and pulmonary risk, function and falls, frailty

score, nutrition, medication review, goals, social support, tests. Only 2 are organ systems; the rest are geriatric domains routine work-up misses.
- **CGA**: multidisciplinary, geriatrician-led — frailty, physical performance (balance, chair rises, timed turn), albumin, creatinine, brain natriuretic peptide, forced expiratory volume in 1 second (FEV1).

3 · GERIATRIC SYNDROMES

- **Frailty, depression, delirium, falls, incontinence, polypharmacy, vision and hearing loss, pressure ulcers.** They feed one another — frailty ↔ urinary incontinence (UI) ↔ falls (rushing to the toilet) — so treating one often improves others.

Frailty

Frailty = decreased reserve → vulnerability to stressors; distinct from comorbidity (coexisting diseases) and disability (inability to self-care).

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) unintentional weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**
Validity	Does not predict postoperative outcome — counting ignores interaction and severity	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm comes with age and interactions, often via the **prescribing cascade** (a side effect treated with another drug). AGS Beers criteria flag inappropriate medicines:

Drug	Concern → action
Antimuscarinics (in dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention → avoid combinations; no tricyclics in benign prostatic hyperplasia (BPH)
Non-steroidal anti-inflammatory drugs (NSAIDs)	Gastrointestinal bleeding; renal and heart failure at creatinine clearance (CrCl) <30 mL/min
Systemic oestrogen; testosterone	Cancer, cardiovascular harm → **vaginal oestrogen is safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), trimethoprim–sulfamethoxazole	Warfarin potentiation → monitor international normalised ratio (INR); dose-reduce at CrCl <30
Benzodiazepines	Cognitive impairment, unsteady gait, delirium → avoid with opioids or ≥2 other central nervous system drugs
Nitrofurantoin	Listed at CrCl <30/chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia → treat nocturia's cause; check sodium

> **KEY POINT** Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- Term coined 2014; a common cause of transient incontinence. More than atrophy: dryness, dyspareunia, urinary tract infection (UTI) **plus urgency and urgency urinary incontinence**. Sex-steroid loss drives overactive bladder (OAB) in both sexes — including men on androgen deprivation therapy or 5α-reductase inhibitors.
- Treat with **vaginal oestrogen** (alternatives: moisturisers, dehydroepiandrosterone, ospemifene). Previous breast/endometrial cancer is not an absolute bar — consult the oncologist.
- **Urethral caruncle**: painless spotting mimicking postmenopausal bleeding — exclude endometrial/cervical cancer. Topical oestrogen → involution; excision rarely needed.

- Faecal incontinence: most dreaded, under-reported — **ask specifically** about bowel function.

4 · INCONTINENCE MANAGEMENT

- Common but not normal; behavioural therapy first → *Voiding Dysfunction book, Ch 110*. †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.
- **Antimuscarinics**: fewer than half persist at 6 months; the ageing blood–brain barrier is more permeable → cognitive harm. Persistence is better with M3-selective agents and informed patients. Retention is uncommon, but check post-void residual after starting and never exceed the maximum dose.

Agent	Point in older adults
Trospium	Quaternary amine → least central penetration
Solifenacin, darifenacin	M3-selective; better persistence
Tolterodine, high dose	QT prolongation → torsades de pointes
Oxybutynin, fesoterodine	Beers "avoid" in dementia
Mirabegron (β3-agonist)	No anticholinergic effect; slight blood pressure rise; metoprolol interaction

Nursing homes

- UI affects three-quarters of long-stay residents and drives admission. Catheter colonisation reaches **98%**, so treating asymptomatic bacteriuria (ASB) causes harm; criteria guide antibiotics:

Loeb — catheter	Loeb — no catheter	National Healthcare Safety Network
≥1: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever + new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 colony-forming units + no other source + clinical signs

> **KEY POINT** A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI AND ASYMPTOMATIC BACTERIURIA

- First separate symptomatic UTI from OAB and ASB. Older adults present atypically, but a **causal link between confusion and UTI is not supported** → *Infections book, Ch 26*.
- ASB: positive culture, regardless of pyuria, without symptoms (up to 22% of older women, ~10% of older men). It raises inflammatory cytokines yet needs no antibiotics; no evidence of graft harm in transplant recipients. US Preventive Services Task Force (USPSTF) grade D against screening; Infectious Diseases Society of America: treat **only in pregnancy and before urological instrumentation**.

6 · CANCER, END OF LIFE

- **Prostate**: USPSTF grade D — no prostate-specific antigen (PSA) screening >70; life expectancy **<10 years** → non-curative plan.
- **Bladder**: restaging resection for T1 disease is often omitted in the elderly yet improves survival. Bacille Calmette–Guérin is less effective (immunosenescence). Cystectomy at ≥75 gives comparable outcomes with careful care.
- Urologists are mandatory reporters of **elder mistreatment** in most places.
- **Palliative care is not withdrawal of care**: aggressive symptom control; goals-of-care conversation first, as family goals may differ.
- Age-Friendly "4 Ms": Mobility, Medication, Mentation, what Matters.

> **KEY POINT** Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ASPREE	Randomised controlled trial (RCT), aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Prevention that works in older adults: calcium + vitamin D (fewer fractures), antihypertensives (lower mortality even >80), statins (less coronary disease), exercise (mobility, independence).

8 · MCQ ONE-LINERS

1. "Geriatrics" coined clinically by Nascher, 1909.
2. Hayflick limit: telomere-driven replicative arrest.
3. Homeostenosis: narrowing of physiological reserve.
4. First BADL lost: bathing; last: feeding.
5. Surgical risk: operative time, ASA, urgency — not age.
6. Fried: ≥3 of 5 frail; 1–2 pre-frail.
7. Deficit-count index does not predict postoperative outcome.
8. NSQIP/AGS checklist: 13 items, 2 organ-system.
9. Trospium: quaternary amine.
0. High-dose tolterodine: QT prolongation.
1. Mirabegron interacts with metoprolol.
2. Azithromycin: Beers-exempt macrolide.
3. Catheter colonisation: up to 98%.
4. Loeb catheter criteria include mental status change.
15. ASB treated only in pregnancy or before instrumentation.
16. USPSTF grade D: ASB screening and PSA >70.
17. Life expectancy <10 years: non-curative prostate plan.
18. Urethral caruncle: topical oestrogen.
19. ASPREE: aspirin increased mortality in healthy elderly.
20. 4 Ms: Mobility, Medication, Mentation, Matters.
21. SASP spreads senescence to neighbouring tissue.
22. GSM causes urgency and urgency incontinence, not only atrophy.
23. Bacillus Calmette–Guérin (BCG) less effective in elderly: immunosenescence.
24. Confusion alone does not diagnose UTI in older adults.

126 Management and Considerations in Transgender and Nonbinary Individuals

Reading time ≈ 22 min (slow pace)

1 · DEFINITIONS AND TERMINOLOGY

- **Transgender and gender-diverse (TGD)**: umbrella term for people whose gender identity differs from the sex assigned at birth. Respectful, non-pathologising language builds trust.
- Four **independent axes** — assigned sex, gender identity, gender expression, sexual orientation. Most terminology errors collapse one axis into another.

Term	Meaning
Gender identity / expression	Deeply felt internal sense of one's gender / its outward display (clothing, name, voice)
Assigned female/male at birth (AFAB/AMAB)	Assignment from external genitalia and/or genetics
Cisgender	Identity congruent with assigned sex
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary; agender; gender fluid	Neither male nor female; no gender; identity changes over time
Gender dysphoria	Distress from the discrepancy between identity and assigned sex
Gender incongruence	Diagnostic term: marked, persistent incompatibility between identity and assigned sex
Sexual orientation	Pattern of attraction — cannot be inferred from gender identity
"Transsexual"	Historical term; disfavoured in many settings

- Pronouns include they/them and neopronouns (ze/hir, xe/xem). Not all TGD people have dysphoria; some need only social transition, others gender-affirming hormone therapy (GAHT) and/or gender-affirming surgery (GAS).

2 · DIAGNOSIS AND ASSESSMENT

	Diagnostic and Statistical Manual of Mental Disorders (DSM-5) (2013)	International Classification of Diseases (ICD-11) (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced ICD-10 "gender identity disorder")
Criteria	Marked incongruence ≥**6 months**, clinically significant distress or impairment, not solely due to prejudice	Marked, persistent incongruence; desire for characteristics and treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

(DSM-5, Diagnostic and Statistical Manual of Mental Disorders, 5th edition; ICD-11, International Classification of Diseases, 11th revision.) ICD-11 keeps a code for access but drops the mental-disorder label to reduce stigma.

- Assessment by a competent professional: gender history and goals, medical and mental health history, screening for conditions that mimic or worsen dysphoria.
- **Severe distress is not a prerequisite** — marked, sustained incongruence is.

World Professional Association for Transgender Health (WPATH) Standards of Care, version 8 (SOC-8, 2022)

- Treat only marked, sustained incongruence; exclude other causes; optimise coexisting conditions.
- Assess **capacity to consent** and understanding of effects on reproduction; **fertility counselling before starting any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy, **consider ≥6 months of hormones** — unless hormones are not clinically indicated (a consideration, not an absolute rule).

> **KEY POINT** SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling remain firm because their consequences are irreversible.

3 · GENDER-AFFIRMING HORMONE THERAPY

- Under supervision safe and effective (WPATH, Endocrine Society). Aim: **physiological** sex-steroid levels of the affirmed gender.
- Eligibility: persistent documented dysphoria, capacity, age of majority, controlled comorbidities.

Direction	Agent	Dose
Transfeminine — oestrogen	Oral oestradiol	2–6 mg/day
	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate intramuscular	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine — antiandrogen	Spironolactone	100–300 mg/day
	Cyproterone acetate	25–50 mg/day
	Gonadotrophin-releasing hormone (GnRH) agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine — testosterone	Enanthate/cypionate	100–200 mg 2-weekly
	Undecanoate	1000 mg 12-weekly (rare in US: oil pulmonary embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

> **KEY POINT** **Why the asymmetry?** Physiological-dose oestrogen alone does not suppress testosterone into the female range, so transfeminine therapy needs two components; testosterone suppresses the ovarian axis unaided, so transmasculine therapy needs one.

Transfeminine regimen

- **Ethinyl oestradiol is not recommended** — ~20-fold thromboembolic risk and cannot be monitored on assays. Commonest: oral 17β-oestradiol, target **100–200 pg/mL**. **Transdermal if venous thromboembolism (VTE) risk**: age >45 or previous VTE.
- **Spironolactone** (US commonest): androgen-receptor antagonist plus partial 17α-hydroxylase/17,20-lyase inhibition. **Cyproterone**: European standard. Alternatives: GnRH agonists, bicalutamide.
- 5α-reductase inhibitors: weaker (testosterone production and binding unaffected); for spironolactone intolerance, partial feminisation or residual hair loss. Progesterone: unproven.
- Effects in 3–12 months; later prostatic/testicular atrophy. **Irreversible: breast development, reduced or absent fertility.**
- Risks (as cisgender hormone replacement): thromboembolism, cardiovascular disease, hypertriglyceridaemia, liver dysfunction, osteoporosis, hormone-related cancer — from over- or under-dosing.
- Monitoring: levels 3-monthly in year 1, then 1–2×/year; target testosterone **<50 ng/dL**; electrolytes/creatinine on spironolactone; prolactin (prolactinoma risk); bone density if risk factors;

stop smoking.

- **After gonadectomy: reduce oestrogen by 50–75%** (titrate to luteinising hormone [LH]/follicle-stimulating hormone [FSH], keep bone density) and **stop the antiandrogen**.

Transmasculine regimen

- Contraindications: pregnancy, unstable coronary disease, untreated polycythaemia. Oestrogen-sensitive cancer → oncology referral (testosterone aromatises). **Contraception essential** — testosterone is teratogenic.
- Target testosterone **300–1000 ng/dL**. Persistent menses → progestin or endometrial ablation; GnRH analogues possible but costly.
- Effects: 1–2 months — menses stop, acne, clitoromegaly, vaginal atrophy, higher libido; later facial/body hair, muscle, deeper voice. **Irreversible: clitoral growth, facial hair, voice change, male-pattern baldness.** Menses return within 2–6 months of stopping.
- Risks: **erythrocytosis**, liver dysfunction, sleep apnoea, cardiovascular disease. Monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2×/year; lipids, blood pressure. Bone density only if testosterone stopped or risk factors.
- **After gonadectomy: no testosterone dose reduction**; progesterone may be stopped.

Nonbinary individuals

- Individualised: AFAB wanting menstrual cessation with partial masculinisation — progestins/intrauterine device/hysterectomy plus testosterone and/or **nandrolone** (muscle without much hair). AMAB wanting feminisation with preserved erections — oestrogen alone or with low-dose antiandrogen.

4 · UROLOGICAL CONSIDERATIONS

Lower urinary tract symptoms (LUTS) in TGD people arise from five contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction, and psychosocial/behavioural factors**.

- **Testosterone-induced vaginal atrophy** (like postmenopausal change): dryness, bleeding, dyspareunia, urgency, frequency → topical vaginal oestrogen (cream, ring, tablet).
- **After penile inversion vaginoplasty**: ~a third report altered voiding (nocturia commonest); spraying/stream divergence from labial asymmetry, scarring or residual corporal tissue; incontinence usually self-limiting; meatal stenosis, stricture; urethroneovaginal fistula from unrecognised urethral injury → *Ch 127*.
- **After phalloplasty/metoidioplasty**: post-void dribbling from urine pooling at the native urethra–pars fixa junction (manual compression at the scrotal base empties it); urethral lengthening makes stricture and fistula far commoner than after vaginoplasty → *Ch 128*.
- **Pelvic floor**: assess and treat dysfunction before surgery; preoperative pelvic floor therapy eases neovaginal dilation.
- **Tucking** (testes into inguinal canals, penis drawn back, held by tape or a gaff): stasis, prostatitis-like symptoms, epididymo-orchitis, urinary tract infection (UTI) from meatal compression near the anus.
- Work-up: urinalysis, uroflowmetry, residual; cystoscopy, imaging, urodynamics as indicated. Exclude atrophy, tucking and pelvic floor causes before blaming the operation.

5 · FERTILITY PRESERVATION

- Counsel **before hormones and/or surgery** (WPATH, American College of Obstetricians and Gynecologists, American Society for Reproductive Medicine, Endocrine Society). Many start GAHT as adolescents, when fertility seems remote.
- Barriers: cost, urgency to transition, provider bias, process-provoked dysphoria.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation** — simple, cheap; hold GAHT ≥**3 months** before collection (GAHT degrades semen)	**Oocyte or embryo cryopreservation** after controlled ovarian hyperstimulation; oocytes avoid committing to a sperm source
Prepubertal / no GAHT pause	Testicular tissue cryopreservation (experimental)	Ovarian tissue cryopreservation — no hormone pause or stimulation needed
Obstacles	Ejaculation may distress; antiandrogen effects → vibratory/electroejaculation or surgical retrieval	Stopping testosterone brings back bleeding; pelvic examination and transvaginal scans distress

After stopping testosterone, menses return in 3–6 months and most conceive within 6 months; testosterone must be stopped when trying to conceive (teratogenic).

6 · CANCER SCREENING — SCREEN THE ORGAN, NOT THE GENDER

- Retained natal organs keep their cancer risk; GAHT's long-term effect is unknown. Smoking, human immunodeficiency virus (HIV) and human papillomavirus (HPV) add risk; access barriers cause later-stage diagnosis.
- **Prostate (transfeminine)**: rare but may be aggressive, mostly when GAHT began after 50. No dedicated guideline — shared decision-making on prostate-specific antigen (PSA) at 55–69. **On GAHT, upper limit 1 ng/mL or rise ≥0.3 ng/mL from nadir**. †Androgen suppression lowers PSA, hence the lower threshold.
- Examine by anatomy: digital rectal examination (natal anatomy or vulvoplasty); after full-depth vaginoplasty the prostate is felt through the **anterior neovaginal wall**. Biopsy transrectal, perineal or transneovaginal.
- **Testis**: only case reports — self-examination only.
- **Cervix (transmasculine)**: screen anyone with a cervix — cytology 3-yearly at 21–29; at 30–65 cytology 3-yearly or high-risk HPV testing/co-testing 5-yearly; stop when no cervix or >65 with two negatives. Smallest speculum, patient-controlled insertion, **vaginal oestradiol 1–2 weeks beforehand** for atrophy.

7 · FOLLOW-UP AND AFFIRMING PRACTICE

- Lifelong organ-appropriate screening, hormone and surgical follow-up; educate patients about their own anatomy because records fragment.
- **Trauma-informed care** (Substance Abuse and Mental Health Services Administration, SAMHSA): safety, trust, collaboration, cultural sensitivity; "Four Rs" — **Realise, Recognise, Respond, Resist retraumatisation**.
- Multidisciplinary team; trained front-desk staff; inclusive intake forms (identity, name, pronouns); a **detailed organ inventory** guides history, examination and screening.
- Examination: support person, explanation, preferred position, warning before touch.

8 · MCQ ONE-LINERS

1. ICD-11 moved gender incongruence to Conditions Related to Sexual Health.
2. DSM-5 gender dysphoria: ≥6 months' incongruence plus distress.
3. WPATH SOC-8 (2022): single professional opinion suffices.
4. Fertility counselling precedes any treatment, not only surgery.
5. Ethinyl oestradiol: not recommended — thrombosis.
6. Transfeminine targets: oestradiol 100–200 pg/mL, testosterone <50 ng/dL.
7. After gonadectomy: oestrogen cut 50–75%, antiandrogen stopped.
8. After gonadectomy: testosterone dose unchanged.
9. Transmasculine target testosterone: 300–1000 ng/dL.
10. Monitor haematocrit on testosterone — erythrocytosis.

1. Irreversible on testosterone: clitoral growth, voice, facial hair, baldness.
2. Post-phalloplasty dribble: pooling at native urethra–pars fixa junction.
3. Hold GAHT ≥3 months before sperm banking.
4. Ovarian tissue freezing needs no GAHT pause.
5. PSA on GAHT: abnormal >1 ng/mL.
16. Prostate after vaginoplasty: palpated via anterior neovaginal wall.
17. SAMHSA Four Rs: Realise, Recognise, Respond, Resist.
18. Testis cancer in transfeminine people: self-examination only.
19. Transdermal oestradiol if age >45 or previous VTE.
20. Neovaginal LUTS: nocturia commonest.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

Reading time ≈ 22 min (slow pace)

1 · AIM AND PREPARATION

- Dual objective for transgender women: align anatomy with gender identity **and** preserve physiological function (voiding, sensation, sexual function). General transgender care, terminology and hormone regimens → *Ch 126*.

Mental health assessment

- Two roles: treating dysphoria, and judging **surgical readiness**. Involve early: months of lead time to address housing, employment, support, smoking, weight.
- Psychiatric comorbidity is common; treat before surgery where possible. Any delay is case-by-case, agreed with the patient.
- Key risks: **failure to dilate**; worsening psychiatric prognosis. Where depression and suicidality are driven by dysphoria, surgery is expected to **reduce** distress — symptoms alone are not a reason to defer.

World Professional Association for Transgender Health (WPATH) Standards of Care version 8 (SOC8)

Older criteria	SOC8 (informed-consent approach)
Two assessments	**One** (high inter-assessor agreement)
Licensed mental health professional only	Any licensed clinician with ≥ master's degree, competent in transgender health, with mental health evaluation in scope — incl. primary care or surgeon
Hormone therapy before gonadectomy	**6 months** of hormone therapy suffices (patient experiences testosterone blockade before an irreversible change)
Comorbidity "well managed"	Risks and benefits identified and discussed
12 months' "real-life experience"	**Removed** — dangerous for those facing violence; social transition discussed instead

- **Capacity is procedure-specific**; if doubtful, a psychologist plans support, possibly with proxy decision-makers.

> **KEY POINT** SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

Care coordination

- Postoperative depression undermines dilation; confirm support, care plan, emergency access.
- Stay within **1 hour's drive** of the hospital for the first **3 weeks**.
- Record must carry affirmed gender and name (misgendering causes dissatisfaction); surgeon confirms WPATH criteria and pre-authorisation.

Body mass index (BMI), hair, hormones

- BMI is the main factor in wound healing and closure tension. No proven cut-off; many require **≤35 kg/m²**; higher may be acceptable for vulvoplasty.
- **Permanent preoperative hair removal** (laser/electrolysis) of the scrotal–perineal template that becomes the canal graft. Intraoperative follicle removal alone fails because hair grows in phases. Canal hair → infection, hairballs.
- **Continue oestrogen perioperatively**: a 2022 meta-analysis showed no added venous thromboembolism (VTE) risk, and stopping causes distress.

Orchiectomy

- Sole procedure or stage one: relieves testis-related dysphoria, eases tucking; afterwards essentially all stop spironolactone, most stop progesterone.
- **Fertility counselling and sperm-banking referral for all**; state clearly that orchiectomy/vaginoplasty cause infertility.
- **Bilateral orchiectomy via a vertical midline scrotal incision**: an inguinal incision may compromise the penile flap's blood supply, while a scrotal incision is incorporated into the later graft.

> **KEY POINT** Plan every orchiectomy as though vaginoplasty will follow — this dictates the midline scrotal incision and leaving peri-cord fat for labial bulk.

2 · PENILE INVERSION VAGINOPLASTY (PIV) — PERINEAL APPROACH

- Commonest technique. Set-up: dorsal lithotomy, pad lateral calf and heel (**common peroneal nerve**); robotic ports as for prostatectomy, steep Trendelenburg; weight-dosed subcutaneous heparin + sequential compression; bleeding is front-loaded → **tranexamic acid 1 g intravenously at start**.

Natal structure	Neo-structure
Glans on dorsal neurovascular bundle	Clitoris
Preputial mucosal collar	Clitoral hood, medial labia minora
Dorsal bulbar urethral plate (3 cm)	Vestibule
Penile shaft skin (inverted)	Canal lining
Scrotal skin (full-thickness graft)	Canal lining (proximal augmentation)
Lateral scrotal skin + peri-cord fat	Labia majora

ALGORITHM
Foley, drain, plug → measure; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion) → harvest scrotal graft + small posterior rhomboid flap (relieves posterior introital tension) → defat, fulgurate follicles, tubularise over dilator → orchiectomy, leave cord fat → penile disassembly → corporal excision + clitoroplasty → urethroplasty → canal dissection → line canal → labiaplasty → pack

- **Disassembly**: circumcising incision sparing the mucosal collar; elevate skin off Buck's fascia **keeping dartos with the skin** (preserves flap blood supply) → cylindrical penile skin flap.
- **Neurovascular bundle** — two options: separate it from tunica, or **keep it on the dorsal tunica** (preferred: less handling, less nerve damage) → open corpora ventrally, excavate spongy tissue, coagulate cavernosal arteries.
- **Excise all erectile tissue** — remnants engorge with arousal and obstruct the canal.
- **Clitoroplasty**: "W" on proximal glans → excise ventromedial glans, keep lateral wings → de-epithelialise and join. Too much glans → prominent, penis-like clitoris. Bundle accordioned onto suprapubic fascia.
- **Urethroplasty**: discard pendulous and distal bulbar urethra above adductor level; excise bulbospongiosus (debulks introitus) **only after canal dissection** — it is the cover for a rectal injury. Spatulate ventrally/downward for a straight seated stream; confirm trajectory with catheter or bougie.

Vulvoplasty (zero-depth) decision

- Vulva without canal — for patients certain they do not want receptive intercourse; avoids **lifelong dilation**, rectal/bladder injury and abdominal access.
- Not freely reversible: scrotal graft discarded; later depth faces scarring.

- **Prior radical prostatectomy → vulvoplasty** (canal dissection may injure the external sphincter → incontinence).

Canal dissection and lining

- Divide central tendon → dissect along urethra toward prostate; Lowsley retractor helps. Repeated **digital rectal examination** (rectal probe cover secured to drape) checks tissue thickness.
- **Preserve Denonvilliers fascia**; if breached (perirectal fat), close it, especially if rectal wall is thin.
- Prostatic capsule breach → check urethra; expect bleeding.
- Lining: tubularised scrotal graft sewn to distal penile skin over dilator; slit inverted skin to deliver clitoris and urethra. **Graft–bed contact by packing** is essential for take.
- Closure: labia majora from lateral scrotal skin with cord fat for bulk; keep incision **≥1 cm** from groin crease (dehiscence). Antibiotic-soaked pack + negative-pressure dressing.

3 · ABDOMINOPERINEAL AND INTESTINAL OPTIONS

- **Peritoneal (Davydov-type) vaginoplasty**: external steps identical; proximal canal made robotically. Rationale: historical rectal injury **~7%** (poor exposure), reduced to ~2.3% (urethral 1.1%).
- Robotic tips: track vasa medially to prostate base; **rectum may mimic seminal vesicles**; minimal cautery; enter between Denonvilliers leaflets (bloodless plane).
- Flaps: **two-flap** (posterior bladder and lateral sigmoid peritoneum, each 10 × 10 cm) or **pull-through** (anterior abdominal wall flap 11 × 20 cm on deep inferior epigastric arteries). **Close peritoneal defects** (hernia).
- **Intestinal (colon/small bowel) vaginoplasty**: for severe penoscrotal hypoplasia or canal stenosis at revision. Contraindications: inflammatory bowel disease, adhesions, irradiation, short gut. Cons: organ injury, mucus, diversion colitis, prolapse; ~1 in 8 removed. **Dilation still required.**

> **KEY POINT** Lining ladder: penile/scrotal skin → peritoneum (more depth or safer exposure) → bowel (no skin left). Each step adds depth and lubrication but imports donor-organ pathology; none removes the duty to dilate.

4 · COMPLICATIONS AND MANAGEMENT

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care, secondary intention; granulation → curette/cautery + topical steroid
Rectovaginal fistula (most devastating)	Rectal injury; **0.8%** primary, 6.3% revision	Intra-op: watertight closure + bulbospongiosus flap ± faecal diversion. Fistula: stop dilating, let canal stenose, revise later (may reopen) **or** delay closure until canal stable
Urethrovaginal fistula (0.8–3.9%)	Missed urethral injury, distal stenosis	Distal (vaginal voiding) often left; bladder-neck (incontinence) → transvaginal repair; Martius flap unusable, gracilis may obstruct introitus
Urethral stricture (1–6%)	Weak/split stream, spraying, urinary tract infection (UTI), retention	Exclude pelvic floor dysfunction; proximal → cystoscopy; dilation for short, else meatal spatulation
Urethral bulge / upward meatus	Under-debulked spongiosum; poor mobilisation to crura	Debulk, fix meatus downward
Vaginal stenosis	Poor dilation, scar, poor graft take, pelvic floor dysfunction	Pelvic floor physiotherapy **first** → scar excision + graft, or peritoneal/bowel augmentation

Complication	Cause / key fact	Management
Prolapse (4–10%)	Poor graft take/fixation, not pelvic floor weakness	No standard; early repack — stenosis often follows. Peritoneal fixation acts as colpopexy

KEY POINT Three failure modes: a violated plane (fistulas), tissue inadequately excised (bulge, engorgement), or a graft not taken/maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient — hence dilation is part of the operation.

5 · OUTCOMES, POSTOPERATIVE CARE, FOLLOW-UP

- Most achieve orgasm; **median time to orgasm ≈ 180 days**. Depth 12–16 cm (cisgender 9–12 cm). Self-lubrication is not the norm.
- Stay ~**5 days**; **day 5: remove pack and catheter**, teach dilation and douching. Retention → recatheterise, remove at 1 week.
- Pelvic floor physiotherapy from before surgery.
- Speculum for depth and granulation (curette; persistent → steroid on dilator tip; odour → antibiotic gel).
- Schedule: weekly × 3, then 6 weeks, 3, 6, 12 months, annually.
- †Prostate is retained — prostate cancer surveillance continues, examined through the anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if depth/calibre stable

KEY POINT The body treats the canal as a defect to close — lifelong stenosis risk. Any narrowing → increase dilation, contact surgeon, physiotherapy.

6 · MCQ ONE-LINERS

1. SOC8: one assessment; 6 months' hormones before gonadectomy.
2. SOC8 removed 12-month real-life experience requirement.
3. Dysphoria-driven depression is not a reason to defer surgery.
4. Main psychosocial surgical risk: failure to dilate.
5. Oestrogen continued perioperatively — no added VTE risk.
6. Orchiectomy incision: vertical midline scrotal.
7. Tranexamic acid 1 g intravenously at start: front-loaded bleeding.
8. Clitoral position landmark: adductor longus tendons.
9. Short penile shaft shortens labia minora, not canal.
10. Residual erectile tissue causes arousal engorgement and canal obstruction.
11. Bulbospongiosus kept until canal done — covers rectal injury.
12. Prior radical prostatectomy favours vulvoplasty.
13. Robotic peritoneal approach: rectum may mimic seminal vesicles.
14. Rectal injury ~7% historically, ~2.3% abdominoperineal.
15. Intestinal vaginoplasty: penoscrotal hypoplasia or revision stenosis.
16. Commonest minor complication: introital wound dehiscence.
17. Rectovaginal fistula: 0.8% primary, 6.3% revision.
18. Bladder-neck urethrovaginal fistula → incontinence; distal → vaginal voiding.
19. Stenosis: pelvic floor physiotherapy first.
20. Neovaginal prolapse reflects poor graft take.
21. Median time to orgasm ≈ 180 days.
22. Dilation 4 × daily for first 6 weeks.

128 Special Urologic Considerations in Transmasculine Individuals

Reading time ≈ 16 min (slow pace)

1 · DEFINITIONS AND SCOPE

- **Gender dysphoria (GD)**: distress from mismatch between gender identity and sex assigned at birth. Older diagnostic labels: transsexualism (International Classification of Diseases, 10th edition [ICD-10]); gender identity disorder (Diagnostic and Statistical Manual of Mental Disorders, 4th edition [DSM-IV]); DSM-5 (2013): gender dysphoria. †ICD-11: "gender incongruence", outside the mental disorders chapter. Terminology, World Professional Association for Transgender Health (WPATH) standards and hormones → *Ch 126*.
- Gender-confirming surgery (GCS) may include mastectomy, hysterectomy–oophorectomy, metoidioplasty, phalloplasty, testicular/erectile prostheses — genital surgery inevitably affects the urinary tract.
- **Youth**: bimodal presentation (childhood, around puberty); high depression, anxiety, self-harm risk. Avoiding public toilets may cause an underactive detrusor. **Early puberty blockade has a surgical price**: an oestrogen-naive vagina makes vaginectomy harder, and underdeveloped labia majora leave less scrotal tissue.
- Care is patient-led, chosen from a "menu"; common goals: male genital appearance, **voiding standing**, erotic sensation, penetration. A dedicated urologist in the multidisciplinary team (MDT) is advisable.

2 · PREOPERATIVE CARE AND CONSENT

- Urethral lengthening, vaginectomy and pelvic floor reconstruction all impair voiding → **mandatory urological screening**: history, uroflowmetry + post-void residual (PVR), renal ultrasound. Treat lower urinary tract dysfunction (LUTD), stones or hydronephrosis first — far harder after urethral lengthening.
- WPATH (Standards of Care version 7): ≥**12 months** continuous testosterone before genital GCS († SOC8, 2022, lowered this to 6 months) — mainly for **clitoral hypertrophy** (growth plateaus ≈2 years).

Agent	Helps surgery	Hinders surgery
Testosterone	Clitoral hypertrophy → tissue for metoidioplasty, urethral lengthening	Vaginal atrophy → harder vaginectomy
Gonadotrophin-releasing hormone (GnRH) agonist	Prevents irreversible female characteristics	Small labia majora → less scrotoplasty tissue

- Consent: urethral fistula/stricture (lengthening); bleeding, mucocele, rectal/bladder/sphincter injury (vaginectomy); LUTD, urinary tract infection (UTI); implant complications.

3 · REMOVAL OF FEMALE STRUCTURES

- Subcutaneous mastectomy usually first. Hysterectomy–oophorectomy preferably laparoscopic (else Pfannenstiel). Ureter runs beneath the uterine artery → injury needs immediate urological repair; bladder injury → immediate repair.
- Remove the **whole cervix and proximal third of vagina** to ease later perineal vaginectomy. †If the cervix is retained, cervical screening continues.
- **Vaginectomy**: hazardous — haemorrhage > bladder injury > rectal injury. Optional, but **no vaginectomy → no urethral lengthening** (high urethrovaginal fistula risk). Alternative: leave upper vagina and fulgurate mucosa (prevents mucocele).

ALGORITHM
Hydrodissect urethrovaginal plane with dilute lidocaine → circumferential introital incision, 4 clamps → blunt rectovaginal plane, gauze protects rectum → **sharp urethrovaginal dissection, no diathermy** (sphincter) → bladder plane (submucosal); bladder injury → close, cystoscopy if ureteric orifices in doubt → lateral walls: free levators, cautery/3-0 sutures for paravaginal plexus → approximate levators (prevents prolapse) → bulbospongiosus over fixed urethra → layered closure

Timing strategy	Verdict
All-in-one	More major complications and transfusion with phalloplasty; none extra with metoidioplasty
Early vaginectomy (with hysterectomy or metoidioplasty), phalloplasty later	Keeps blood loss and hypotension away from free-flap perfusion (theoretical)
Two-stage (preferred): mastectomy + hysterectomy–oophorectomy; then vaginectomy + genitoperineal reconstruction	Authors' choice

4 · GENITAL RECONSTRUCTION

Urethral part	Construction	Male equivalent
Native	Original urethra	—
Fixed (perineal)	Vestibular mucosa + inner labia minora, 3-cm plate tubularised over 18-French (Fr) tube	Bulbar, no spongiosum
Phallic	Tube-in-tube skin island (radial forearm free flap) or separate flap	Penile, no spongiosum

- Clitoris de-epithelialised, fixed to pubis; one **dorsal clitoral nerve** (1 or 11 o'clock, on the anastomosis side) coapted to a flap cutaneous nerve for sensation.
- **Bulbospongiosus wrapped over the fixed part**: vascular cover (fewer fistulas) and expulsion of last drops. Anterior vaginal wall flap avoided (random flap, necrosis, sacculation).
- **Scrotoplasty**: midline closure leaves the scrotum between the legs → labial flaps (external pudendal artery) rotated 180° medially to lie in front.
- **Phalloplasty** (usually plastic surgeon): many flaps described — none ideal.
- **Metoidioplasty**: hormone-enlarged clitoris; divide chordee, release labia minora and the tethering ventral urethral plate; lengthen urethra with transverse preputial island or labia minora flap ± buccal mucosa.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · POSTOPERATIVE CARE AND CATHETERISATION

- Suprapubic catheter (SPC) + neourethral catheter; remove neourethral at **14–21 days**; clamp SPC, open after each void to record PVR; keep until voiding normal (oedema, post-vaginectomy neurogenic dysfunction).
- Catheterisation is difficult: abrupt curve into native urethra; Bartholin-duct sinuses catch the tip.

ALGORITHM
One gentle Coudé attempt → stylet-guided attempt → any difficulty: stop → flexible cystoscopy, guidewire, open-tip catheter. Emergency: **suprapubic catheter**

6 · URETHRAL COMPLICATIONS

- **Fistula**: commonest at phallic base. Keep diversion until wound heals; small ones close spontaneously; persisting >**3 months** → excise, layered closure with fat interposition (no dartos), catheter 10–12 days.
- **Stricture**: commonest at the **fixed–phallic anastomosis** (poorly perfused flap end); also meatus. Plateau uroflow → retrograde urethrography (RUG). Treat only after **≥6 months**; retention → SPC or perineal urethrostomy.

Option	Use
Dilation	Low-grade meatal stenosis; not curative for anastomotic stricture
Direct vision internal urethrotomy (DVIU)	Cold knife over guidewire, catheter 2 weeks; low success, never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative strictures
Excision and primary anastomosis	Short, near-obliterative anastomotic strictures
Free graft	Needs flap support (fat bed) or dorsal inlay on clitoral corpus
Two-stage (standard)	Long or complex strictures; retubularise ≥6 months later
Flap / perineal urethrostomy	Recurrent; or no further reconstruction wanted

- **Hair**: prevent with hairless flap or preoperative laser; bezoar/stones → open phallic urethra, remove, epilate, retubularise later.
- **Post-void dribbling**: near-universal (rigid tube, no spongiosum) → perineal pressure and phallic milking; severe → exclude sacculation or vaginal-remnant diverticulum.

KEY POINT The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · PROSTHESES

- Testicular: ≥6 months after scrotoplasty; 17-mL implants, fixed.
- Erectile: ≥**1 year** after phalloplasty, once **protective sensation** present and urethra stable. One cylinder; no tunica → anchor to pubic periosteum (rear tip extender, Dacron sleeve or dedicated device with pubic base plate). Malleable is simpler.
- Infection/erosion frequent → urgent explantation, reimplant ≥6 months.

8 · MCQ ONE-LINERS

1. Testosterone duration before genital GCS: ≥12 months (SOC7); † 6 months (SOC8).
2. Testosterone-induced clitoral growth plateaus at ~2 years.
3. Testosterone helps metoidioplasty but hinders vaginectomy.
4. Puberty blockade reduces scrotoplasty tissue.
5. No vaginectomy → no urethral lengthening.
6. Commonest vaginectomy complication: haemorrhage.
7. Urethrovaginal dissection without diathermy protects sphincter.
8. Neourethra: native, fixed, phallic parts.
9. Commonest stricture site: fixed–phallic anastomosis.
10. Commonest fistula site: phallic base.
11. Fistula >3 months: permanent, repair.
12. Defer stricture treatment ≥6 months post-phalloplasty.
13. Standard for long phallic strictures: two-stage urethroplasty.
14. Post-void dribbling: near-universal.
15. Erectile prosthesis ≥1 year, needs protective sensation.
16. Emergency retention after GCS: suprapubic catheter.

7. Metoidioplasty phallus 3–8 cm, spontaneous erection.
8. Scrotoplasty flaps: external pudendal artery, rotated 180°.
19. Repeated DVIU for anastomotic stricture: futile.
20. Catheterisation sequence: Coudé, stylet, then cystoscopy-guided.
21. Erectile cylinder anchored to pubic periosteum.

★ Putting it together: genitalia, DSD, gender and special populations

Reading time ≈ 8 min (slow pace) · Integrates Ch 24, 25, 55, 58–60, 65, 120, 126–128 · use these as the skeleton of long-essay answers

A · NEWBORN WITH ATYPICAL GENITALIA

ALGORITHM

1. Treat as an emergency for the adrenal, not the genitalia: congenital adrenal hyperplasia (CAH) is the commonest cause (half of all, 90% of 46,XX cases). Salt-wasting crisis (low sodium, high potassium, shock) usually appears at day 5–7, so normal early electrolytes do not exclude it (→ *Ch 59*).

2. Tests: karyotype (with SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound for a uterus; then hormonal and genetic work-up guided by karyotype.

3. Team: † defer sex assignment until evaluated by a multidisciplinary DSD team (endocrinology, urology, genetics, psychology) with the family.

4. Gonads at risk: dysgenetic gonads with Y material (e.g. complete gonadal dysgenesis, 30–45% tumour risk) → prophylactic gonadectomy.

B · UROGENITAL SINUS AND CLOACA: WHAT DECIDES THE OPERATION

KEY POINT True urethral length (bladder neck to confluence) matters more than common-channel length. Low confluence → flap vaginoplasty; high confluence → pull-through or total/partial urogenital mobilisation. Clitoroplasty only for severe hypertrophy (Prader III–IV), sparing the dorsal neurovascular bundle. In cloaca: early colostomy, decompress hydrocolpos and protect the kidneys, then image (cloacogram, MRI) and endoscope before definitive posterior sagittal repair; look for VACTERL associations and tethered cord (→ *Ch 60*).

C · ABSENT VAGINA

Cause	Key feature	Management
Mayer–Rokitansky–Küster–Hauser (MRKH) syndrome	46,XX, normal ovaries and breasts, absent uterus; † image the kidneys (renal anomalies are common)	Dilation first line (needs a dimple and compliance); surgery (McIndoe graft, bowel, buccal mucosa, peritoneal) if it fails
Complete androgen insensitivity	46,XY, testes, no uterus	Vaginal dilation usually suffices (→ *Ch 59*)
Transverse septum / imperforate hymen	Uterus present, haematocolpos at puberty	Imperforate hymen: transverse incision, not aspiration; septum: excision (→ *Ch 58*)

D · IMMUNOTHERAPY AND HEREDITARY CANCER

ALGORITHM

Checkpoints: tumour interferon-γ induces programmed death ligand-1 (PD-L1), which switches off PD-1-positive T cells; cytotoxic T-lymphocyte antigen-4 (CTLA-4) inhibitors act earlier, on priming. † Immune-related adverse events (colitis, hepatitis, pneumonitis, endocrinopathy): hold the drug and give steroids for grade ≥2; endocrinopathies need lifelong replacement (→ *Ch 24*).

Hereditary syndromes to recognise: von Hippel–Lindau (VHL, clear cell renal cancer; belzutifan), hereditary leiomyomatosis and renal cell cancer (fumarate hydratase; aggressive papillary), Birt–Hogg–Dubé (FLCN; hybrid oncocytic tumours), Lynch (mismatch repair; upper-tract urothelial cancer). BRCA/homologous-repair defects in prostate cancer → PARP inhibitor; FGFR alteration in urothelial cancer → erdafitinib (→ *Ch 25*).

E · OLDER PATIENT BEFORE UROLOGICAL SURGERY

KEY POINT Function, not age, predicts outcome. Screen for frailty, then do a comprehensive geriatric assessment if positive: cognition and capacity, delirium risk, nutrition, falls, polypharmacy. Avoid anticholinergic burden (prefer a β3-agonist for overactive bladder). For new incontinence, first look for transient causes († DIAPPERS: delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess urine output, restricted mobility, stool impaction) (→ *Ch 120*).

F · TRANSGENDER AND NONBINARY PATIENTS: UROLOGICAL POINTS

ALGORITHM

Use the patient's name and pronouns; organ inventory guides screening — retained prostate after vaginoplasty (examine through the anterior neovaginal wall; † PSA runs lower on oestrogen), retained cervix after testosterone (→ *Ch 126*).

Vaginoplasty complications: canal stenosis, rectovaginal fistula (most devastating), urethrovaginal fistula (→ *Ch 127*). Phalloplasty: urethral lengthening brings fistula and stricture; skipping vaginectomy raises fistula risk (→ *Ch 128*).